
Convergent or Persistent? An Empirical Taxonomy of Auxiliary Loss Dynamics in Vision MoE

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Abstract

1 Every auxiliary loss in sparse Mixture-of-Experts training imposes a gradient-level
2 tax on router optimization—yet this cost has never been measured. We introduce
3 the **Gradient Interference Matrix (GIM)**, a per-layer diagnostic that decomposes
4 auxiliary–task gradient interactions on router weights into magnitude dominance
5 and directional alignment.
6 Across three datasets, two architectures, and scales from ~ 7 M to ~ 98 M param-
7 eters, GIM reveals a surprising structure: all auxiliary losses converge toward the
8 task gradient, but at strikingly different rates. Load-balancing losses converge
9 slowly with persistent gradients; expert-router alignment losses self-extinguish
10 within tens of epochs—creating an order-of-magnitude mid-training separation.
11 This taxonomy is not fixed: below a critical λ threshold the distinction vanishes,
12 while above it the ordering emerges abruptly (13/13 matched pairs, $p < 0.001$) and
13 grows monotonically.
14 We operationalize these findings through GIM-Switch, an adaptive scheduler that
15 autonomously modulates auxiliary strength via bidirectional tier transitions, and
16 *phantom GIM*, which quantifies the interference that loss-free balancing sidesteps.
17 The auxiliary loss tax is real but manageable—GIM tells practitioners when to act
18 and when to leave well enough alone.

19 1 Introduction

20 Sparse Mixture-of-Experts (MoE) enables efficient scaling by activating a subset of parameters per
21 input, achieving favorable quality–compute trade-offs in language modeling [Shazeer et al., 2017,
22 Fedus et al., 2022, Lepikhin et al., 2021] and, increasingly, in vision [Riquelme et al., 2021, Daxberger
23 et al., 2024]. A learned router assigns each token to k of E experts; preventing router collapse and
24 ensuring balanced utilization requires auxiliary losses—load-balancing [Fedus et al., 2022, Lepikhin
25 et al., 2021], router z-loss [Zoph et al., 2022], expert-coupling penalties [Lv et al., 2025]—that have
26 become a standard, largely unexamined component of MoE training across all scales.

27 These auxiliary objectives are typically viewed as lightweight regularizers, tuned via a single scalar
28 coefficient λ and otherwise left alone. Yet every auxiliary loss injects gradients into the router pa-
29 rameters *at every training step*. Because router weight matrices are small ($E \times D$, e.g. $8 \times 192 = 1,536$
30 parameters per layer) relative to the expert parameters they gate, even modest auxiliary gradients can
31 dominate the task gradient in this low-dimensional subspace—imposing what we term an *auxiliary*
32 *loss tax* on router optimization.

33 The practical relevance of this tax has drawn increasing attention. Loss-free balancing [Wang et al.,
34 2024, DeepSeek-AI, 2024] sidesteps auxiliary losses entirely, explicitly motivated by the desire to
35 avoid “interference gradients that impair model performance.” DirMoE [Vahidi et al., 2026] replaces
36 load-balancing with differentiable Dirichlet routing to eliminate the associated gradient contamination.

STGC [Yang et al., 2025] identifies token-level gradient conflicts within experts. These solutions presuppose the tax is harmful—yet the *magnitude*, *dynamics*, and λ -*dependence* of auxiliary loss gradients on router parameters have never been systematically measured.

Standard multi-task learning (MTL) gradient tools [Chen et al., 2018, Yu et al., 2020, Liu et al., 2021, Navon et al., 2022] decompose conflicts among *distinct tasks* with independent forward passes. MoE auxiliary losses are structurally different: they are regularizers of a *single task*, targeting a shared, low-dimensional parameter set (the router weights) from the *same* forward pass. At **micro scale** ($\approx 5\text{--}30$ M parameters)—increasingly relevant for edge deployment [Daxberger et al., 2024] and academic research [Muennighoff et al., 2025]—the task gradient signal is smaller, further amplifying the relative impact of auxiliary gradients.

We introduce the **Gradient Interference Matrix (GIM)**, a per-layer diagnostic that decomposes pairwise gradient relationships on router parameters into three complementary quantities: the **Gradient Interference Ratio** (GIR, magnitude dominance), **Angular Alignment** (AA, directional conflict), and **Gradient Contribution Fraction** (GCF, λ -weighted effective share). The **effective GIR** (eGIR) $= \lambda \cdot \text{GIR}$ captures the λ -attenuated operational impact, enabling lifecycle diagnosis throughout training.

Across three datasets (CIFAR-100, ImageNet-50, Tiny-ImageNet), two architectures (ViT-MoE, ConvNeXt-MoE), and scales from ~ 7 M to ~ 98 M parameters, GIM reveals that the auxiliary loss tax is *real but structured*. All auxiliary losses converge toward the task gradient, but at strikingly different rates: load-balancing converges slowly with persistent interference, while expert-routing alignment losses self-extinguish within tens of epochs—creating a $4\text{--}12\times$ mid-training GIR separation. This separation is λ -dependent: below a critical threshold the distinction vanishes, while above it the ordering emerges abruptly (13/13 matched pairs at $\lambda \geq 0.1$, $p < 0.001$) and grows monotonically with λ .

We operationalize these findings through **GIM-Switch**, an adaptive λ -scheduler that modulates auxiliary strength based on real-time eGIR signals, and **phantom GIM**, which quantifies the interference that loss-free balancing sidesteps.

Our contributions: (1) The GIM diagnostic framework with *phantom GIM* for loss-free balancing; (2) a λ -dependent convergence spectrum—convergent vs. persistent—validated across three datasets, two architectures, and scales up to ~ 98 M; (3) effective GIR showing the tax is bounded and consistent with routing-distribution reshaping; (4) GIM-Switch, a GIR-derived adaptive λ -scheduler with validated cross-seed consistency.

2 Related Work

Solution-oriented MoE design. Recent work addresses auxiliary loss interference through architectural redesign: DirMoE [Vahidi et al., 2026] replaces load-balancing with differentiable Dirichlet routing; loss-free balancing [Wang et al., 2024, DeepSeek-AI, 2024, Han and Zhong, 2025], Soft MoE [Puigcerver et al., 2024], and ReMoE [Wang et al., 2025] sidestep auxiliary losses entirely. CDSP-MoE [Gan and Lei, 2025] prunes interfering pathways using gradient conflict signals; Lv et al. [2025] propose expert-router coupling enforcing bidirectional alignment. GIM complements these by providing lifecycle-level *diagnosis* of gradient interference—Fedus et al. [2022] establish the standard auxiliary loss formulation that GIM characterizes at the gradient level.

Micro-scale and vision MoE. Most MoE research targets billion-parameter language models [Cai et al., 2024, Du et al., 2022]. At smaller scales, V-MoE [Riquelme et al., 2021] and Mobile V-MoEs [Daxberger et al., 2024] demonstrate MoE viability for vision, while OLMoE [Muennighoff et al., 2025] and MoE++ [Jin et al., 2025] explore small-scale language MoE—none examine auxiliary loss gradient dynamics.

Gradient analysis in MTL and MoE. GradNorm [Chen et al., 2018], PCGrad [Yu et al., 2020], CAGrad [Liu et al., 2021], and Nash-MTL [Navon et al., 2022] manage gradient conflicts among *distinct tasks*; MoE auxiliary losses are regularizers on the *same task* targeting shared router weights. Liu et al. [2024b] compare router architectures; GRIN [Liu et al., 2024a] uses sparse gradient estimation for routing; STGC [Yang et al., 2025] resolves token-level conflicts within experts. GIM

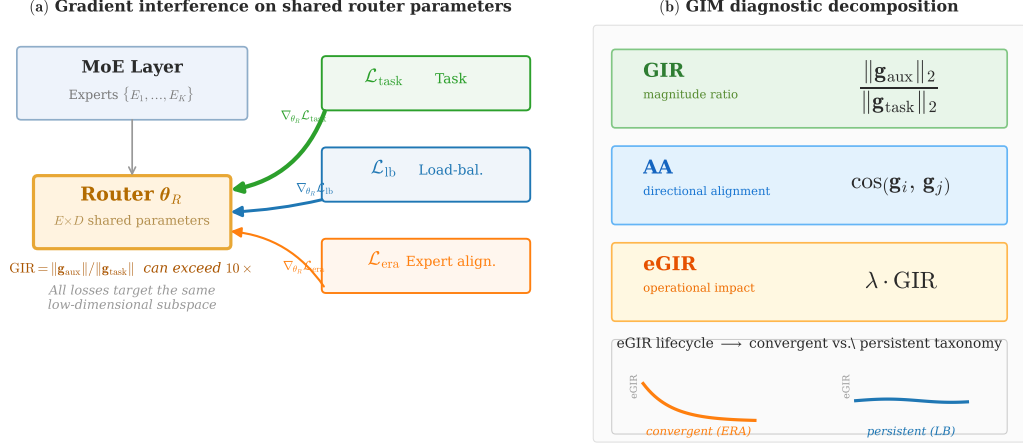


Figure 1: Overview of the GIM diagnostic framework. Given a forward pass through an MoE layer, separate backward passes compute per-loss gradients on router parameters θ_R . GIM extracts three complementary diagnostics per layer: the Gradient Interference Ratio (GIR, magnitude dominance), Angular Alignment (AA, directional conflict), and Gradient Contribution Fraction (GCF, λ -weighted effective share).

88 provides a *systematic diagnostic protocol*: per-layer, router-specific, pairwise, with lifecycle analysis
 89 across the full training trajectory.

90 3 Method

91 We introduce the **Gradient Interference Matrix (GIM)**, a per-layer diagnostic that decomposes
 92 auxiliary loss gradients on router parameters into magnitude ratio (GIR), directional alignment (AA),
 93 and λ -weighted contribution fraction (GCF), as illustrated in Figure 1.

94 3.1 Gradient Interference Matrix

95 Consider a Mixture-of-Experts model with K auxiliary losses $\mathcal{L}_1, \dots, \mathcal{L}_K$ in addition to the task loss
 96 $\mathcal{L}_{\text{task}}$, trained with the combined objective

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \sum_{j=1}^K \lambda_j \mathcal{L}_j, \quad (1)$$

97 where λ_j is the coefficient for auxiliary loss j . Let $\theta_R^{(l)}$ denote the router parameters at MoE layer l ,
 98 and define the per-loss gradient vectors

$$\mathbf{g}_i^{(l)} = \nabla_{\theta_R^{(l)}} \mathcal{L}_i, \quad i \in \{\text{task}, 1, \dots, K\}. \quad (2)$$

99 The **Gradient Interference Matrix** at layer l and training step t is the $(K+1) \times (K+1)$ matrix of
 100 pairwise cosine similarities:

$$\text{GIM}^{(l)}[i, j] = \frac{\mathbf{g}_i^{(l)} \cdot \mathbf{g}_j^{(l)}}{\|\mathbf{g}_i^{(l)}\|_2 \|\mathbf{g}_j^{(l)}\|_2}. \quad (3)$$

101 The off-diagonal entries define the **Angular Alignment (AA)** between loss pairs ($\text{AA} \in [-1, 1]$);
 102 this directional component is essential for interpreting GIR magnitude (Sections 4.1.2–5.2).

103 3.2 Gradient Interference Ratio

104 The **Gradient Interference Ratio (GIR)** quantifies magnitude dominance of auxiliary loss i relative
 105 to the task loss:

$$\text{GIR}_i(l, t) = \frac{\|\mathbf{g}_i^{(l)}\|_2}{\|\mathbf{g}_{\text{task}}^{(l)}\|_2}. \quad (4)$$

A high GIR indicates that the auxiliary gradient is large relative to the task gradient; however, magnitude alone does not determine interference. When paired with near-orthogonal AA ($|\cos| \approx 0$), a high GIR produces a *ghost gradient*—large but directionally inert—whereas non-negligible AA signals genuine competition for the same descent subspace (Section 4.1.2). The accuracy effect of the ghost-gradient regime is architecture-dependent (Appendix E).

The **Gradient Contribution Fraction (GCF)** measures the λ -weighted share of each loss in the combined router gradient:

$$f_i = \frac{\lambda_i \|\mathbf{g}_i^{(l)}\|_2}{\|\mathbf{g}_{\text{task}}^{(l)}\|_2 + \sum_{j=1}^K \lambda_j \|\mathbf{g}_j^{(l)}\|_2}. \quad (5)$$

3.3 Effective GIR

Raw GIR captures magnitude dominance but not the actual contribution to the parameter update, which is attenuated by the loss coefficient λ_i . The **effective GIR (eGIR)** is defined as:

$$\text{eGIR}_i(l, t) = \lambda_i \cdot \text{GIR}_i(l, t), \quad (6)$$

providing the λ -weighted magnitude ratio that governs the auxiliary loss’s actual share of the router gradient update. When $\text{eGIR} \ll 1$, the auxiliary loss contributes negligibly to the combined update even if raw GIR is large.

eGIR enables a *temporal diagnostic*: it distinguishes **convergent** interference (declining eGIR as the auxiliary objective is satisfied, e.g., ERA) from **persistent** interference (stable eGIR throughout training, e.g., load-balancing), informing whether λ management is warranted (Section 5.2).

We additionally define a directional variant (deGIR) in Appendix B.6 and base- λ eGIR for adaptive scheduling in Appendix A.

3.4 GIM-Switch: Adaptive λ Scheduling

GIM-Switch uses a sliding-window smoothed eGIR signal ($\overline{\text{eGIR}}$, window $W=3$; Appendix A) to assign each auxiliary loss to one of three tiers:

$$\lambda_i(t) = \begin{cases} \lambda_i^{(0)} & \text{if } \overline{\text{eGIR}}(t) < \tau_1 \quad (\text{SAFE}), \\ \gamma \cdot \lambda_i^{(0)} & \text{if } \tau_1 \leq \overline{\text{eGIR}}(t) < \tau_2 \quad (\text{MONITOR}), \\ 0 & \text{if } \overline{\text{eGIR}}(t) \geq \tau_2 \quad (\text{INTERVENE}), \end{cases} \quad (7)$$

where $\gamma \in (0, 1)$ is the attenuation factor.

3.5 Experimental Setup

Architecture and training. ViT-Tiny MoE [Dosovitskiy et al., 2021] (6.86M, $D=192$, depth 6, MoE at layers 2/4, $E=8$ experts, top-2 routing, linear routers $\mathbf{W}_R^{(l)} \in \mathbb{R}^{E \times D}$); ViT-Small ($\sim 46\text{M}$; Appendix B); ViT-Base-MoE ($\sim 98\text{M}$; Section 4.2.5). Three auxiliary losses: **LB** [Fedus et al., 2022] ($\lambda=0.01$, persistent), **ERA** ($\lambda \in \{0.025-0.5\}$, convergent), **z-loss** [Zoph et al., 2022] ($\lambda=0.001$, persistent); full definitions in Appendix B. CIFAR-100 (batch 128, AdamW [Loshchilov and Hutter, 2019], 90 epochs, $n=6$); ImageNet-50 (50 classes, 32×32 , $n=5$); Tiny-ImageNet (200 classes, 64×64 , 30 epochs, $n=2-3$); ViT-Base-MoE ($n=2$, 80 epochs).

4 Results

4.1 CIFAR-100

We study CIFAR-100 using a ViT-Tiny MoE (6.86M parameters, MoE at blocks 2 and 4, 8 experts, top-2 routing), with fixed $\lambda_{\text{lb}} = 0.01$ and varying $\lambda_{\text{era}} \in \{0.025-0.5\}$. In 30-epoch sweeps (Appendix C.11), ERA gradient norm exhibits a **self-limiting** $10\times$ decrease across the λ range as higher coefficients accompany faster convergence.

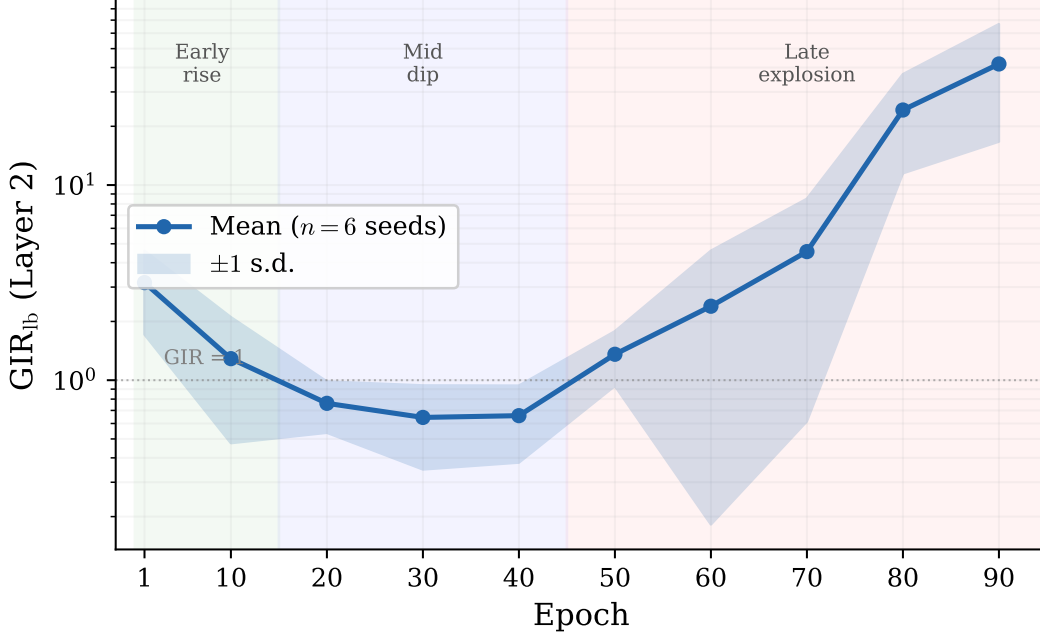


Figure 2: GIR_{lb} trajectory over 90 epochs on CIFAR-100 ViT-Tiny MoE ($\lambda_{\text{era}} = 0.1$). The U-shaped pattern reflects denominator collapse: as task accuracy saturates, the task gradient vanishes while the LB gradient persists.

142 4.1.1 Why Does GIR Explode in Late Training?

143 Load-balancing GIR rises to 16–75 $\times$ by epoch 90 across six seeds—seemingly alarming, but
 144 misleading. This U-shaped trajectory (Figure 2) is denominator-driven: the task gradient collapses
 145 $\sim 130\times$ on CIFAR-100 as training accuracy saturates to $>99.9\%$, while the LB gradient barely
 146 changes. The escalation is regime-specific, absent on ImageNet-50 and Tiny-ImageNet where task
 147 gradients remain substantial (Section 5.1).

148 After λ -weighting ($\lambda_{\text{lb}}=0.01$), effective GIR stays below $0.02\times$ during active learning (epochs 5–50)
 149 and never exceeds $0.75\times$ —the “tax” is bounded.

150 **Differential convergence dynamics.** ERA gradients self-extinguish (*convergent*), LB gradients
 151 persist, and AA oscillates near zero—a **slow-converging vs. fast-converging spectrum** where
 152 interference is magnitude-driven, not directional (Appendix C.5). At $\lambda=0.1$, matched-pair com-
 153 parisons across three seeds show LB produces $3.3\text{--}3.5\times$ higher GIR than ERA, corroborating the
 154 divergent interference profiles underlying the convergence-rate taxonomy. This magnitude ordering
 155 is λ -threshold mediated, emerging above $\lambda \approx 0.1$; at $\lambda_{\text{era}}=0.01$, where ERA operates in the amplifier
 156 regime (Section 4.1.2), the LB–ERA GIR difference is not consistently directional.

157 **Implications.** Despite escalating GIR, GCF analysis (Eq. 5, Appendix C.12) shows the task
 158 gradient retains $\sim 72\%$ of the router update at epoch 90, and multi-seed analysis ($n=6$) finds accuracy
 159 differences within ± 2 pp (Table 1).

160 4.1.2 ERA λ Sensitivity: Stabilizer–Amplifier Transition

161 A 7-point λ_{era} sweep ($n=3$; Table 9 in Appendix C.14) reveals four behavioral regimes. The optimum
 162 at $\lambda=0.2$ ($\Delta = +1.47$ pp, std 0.80 pp $<$ baseline 1.05 pp) defines a *stabilizer regime*. At $\lambda=0.01$,
 163 seed s123 collapses (-3.34 pp)—an *amplifier regime*. At $\lambda=0.001$, GIR explodes ($48\text{--}59\times$) but
 164 $|\cos \theta| < 0.10$: a *ghost gradient* that is directionally inert (Appendix C.14; architecture-dependent
 165 effects in Appendix E).

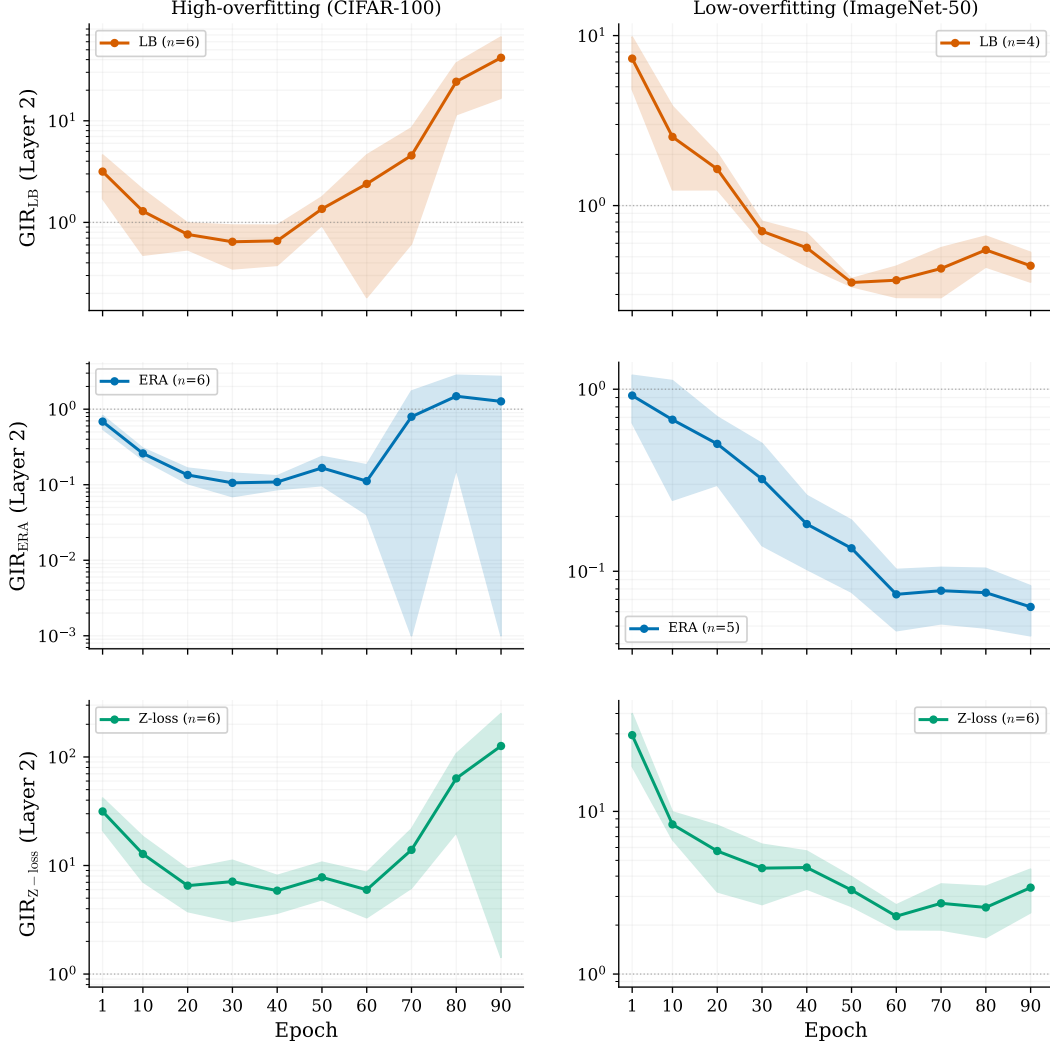


Figure 3: GIR trajectories across regimes (Layer 2). High-overfitting: LB/Z-loss GIR escalates (denominator-driven); ERA $\leq 4\times$. Low-overfitting: all losses stay bounded.

Cross-architecture. ConvNeXt-MoE ($n=3$) shows monotonically increasing $\bar{\Delta}$ with λ (contrasting ViT-Tiny’s inverted-U), confirming architecture-dependent regime boundaries (Appendix C.6).

Z-loss. Router z-loss ($\lambda_z=0.001$, $n=6$) exhibits the highest GIR (2.8–375 $\times$) yet establishes equivalence ($\bar{\Delta}=+0.69$ pp, $p_{\text{Dunn}}=0.045$; Appendix C.16).

4.2 Does the Tax Scale with Dataset Difficulty?

The CIFAR-100 U-shape raises a natural question: does GIR escalation reflect a fundamental property of auxiliary losses, or an artifact of training saturation? A 2×2 ablation—{CIFAR-100 (high-overfitting, >99.9%), ImageNet-50 (low-overfitting, $\Delta_{\text{gap}} \approx 38\text{--}41$ pp)} $\times$ {LB, ERA}—reveals a clean answer.

4.2.1 Load-Balancing in the Low-Overfitting Regime

On ImageNet-50, GIR_{lb} settles at 0.33–0.66 $\times$ by epoch 90 (Figure 3); with $\lambda_{\text{lb}}=0.01$, effective GIR $< 0.006\times$. The divergence is denominator-driven: the task gradient *strengthens* on ImageNet-50 while collapsing $\sim 150\times$ on CIFAR-100 (Appendix D).

Table 1: Regime ablation. GIR at L2, ep90. TOST: $\delta = \pm 2$ pp, Dunnett-corrected ($k=3$, $\rho=0.5$). CIFAR-100: $n=6$; ImageNet-50: $n=5$ (LB/ERA), $n=6$ (z-loss).

Dataset	Aux Loss	GIR (ep90)	Test Acc (%)	$\bar{\Delta}$ (pp)	p_{TOST}	p_{Dunn}
CIFAR-100 (high-overfit)	No-Aux	—	45.3 ± 1.0	—	—	—
	LB	$2.8\text{--}75 \times^\dagger$	45.6 ± 0.7	+0.34	0.012	0.027
	ERA	$0.05\text{--}4.3 \times^\ddagger$	45.5 ± 1.0	+0.25	0.011	0.024
	Z-loss	$2.8\text{--}375 \times^\S$	46.0 ± 0.7	+0.69	0.020	0.045
ImageNet-50 (low-overfit)	No-Aux	—	48.3 ± 1.1	—	—	—
	LB	$0.33\text{--}0.66 \times$	47.8 ± 0.8	−0.46	0.023	0.051
	ERA	$0.04\text{--}0.06 \times$	48.8 ± 0.3	+0.50	0.023	0.051
	Z-loss	$2.2\text{--}4.3 \times$	48.7 ± 0.5	+0.43	0.021	0.047 [¶]

[†] One seed: router collapse (GIR = $2.8 \times$); remaining five: $16\text{--}75 \times$.

[‡] Four seeds: $0.05\text{--}0.78 \times$; two reach $1.7 \times / 4.3 \times$ under strong task gradient collapse.

[§] Full range across $n=6$ seeds; highest value (1337 , $375 \times$) associated with near-complete task gradient collapse at L2.

[¶] ρ -sensitive: passes at $\rho \geq 0.381$; margin 0.0003 at empirical $\bar{\rho}$ (Appendix L).

Table 2: Dose-response on Tiny-ImageNet (unified terminal ratio, epoch 30): taxonomy ratio = mean GIR_{lb} / mean GIR_{erc} from matched-pair comparisons at each λ . Sign test at $\lambda \geq 0.1$: 13/13 pairs show LB > ERA ($p < 0.001$).

λ	Taxonomy Ratio	$\pm$ Std	Seeds	Strength
0.01	$\approx 1.0 \times$	—	3	Null ($p=0.991$)
0.05	$0.983 \times$	0.107	3	Null
0.1	$1.747 \times$	0.401	3	Strong
0.2	$2.321 \times$	0.180	2	Strong
0.3	$2.59 \times$	0.72	2	Strong

4.2.2 The 2×2 Regime–Loss Matrix

Raw GIR escalation requires both a *persistent* auxiliary gradient and a *collapsing* task gradient—only CIFAR-100 $\times$ persistent loss cells exhibit it. Despite alarming raw GIR, accuracy impact is negligible: Dunnett-corrected TOST [Schuirmann, 1987, Lakens, 2017] establishes equivalence for all CIFAR-100 losses ($p_{\text{Dunn}} \leq 0.045$); ImageNet-50 z-loss narrowly passes ($p_{\text{Dunn}}=0.047$, ρ -sensitive); LB/ERA narrowly miss ($p_{\text{Dunn}} \approx 0.051$).

4.2.3 Preliminary Scale Exploration: ViT-Small MoE

Preliminary ViT-Small results (46.2 M, $6.7 \times$ ViT-Tiny) are directionally consistent: LB suppresses expert load variance by 85–87% while z-loss preserves baseline specialization ($n=3\text{--}6$; Appendix F).

4.2.4 Cross-Dataset Dose-Response: Taxonomy Ratio vs. λ on Tiny-ImageNet

The convergence-rate taxonomy established on CIFAR-100 could be dataset-specific. To test generalization, we run matched LB–ERA pairs on Tiny-ImageNet across five λ conditions, asking: at what auxiliary strength does the taxonomy ordering emerge? (Table 2; Figure 4).

The dose-response (Figure 4) reveals two regimes: a **null zone** ($\lambda \leq 0.05$, ratio ≈ 1.0) and a **monotonic separation** above $\lambda \approx 0.1$ (sign test: 13/13 LB > ERA, $p < 0.001$), growing from $1.75 \times$ to $2.59 \times$ with increasing cross-seed variance (± 0.18 at $\lambda=0.2$ to ± 0.72 at $\lambda=0.3$).

4.2.5 Medium-Scale Validation: ViT-Base-MoE (~ 98 M)

ViT-Base-MoE (~ 98 M, $14 \times$ ViT-Tiny, $n=2$ seeds, ImageNet-50) confirms the taxonomy ordering at medium scale: LB GIR is $10\text{--}67 \times$ larger than ERA GIR across all layer $\times$ seed combinations, with LB following a convergent trajectory from $2.1\text{--}3.6 \times$ (epoch 1) to $0.5\text{--}0.8 \times$ (epoch 80; Table 14 in Appendix D.5).

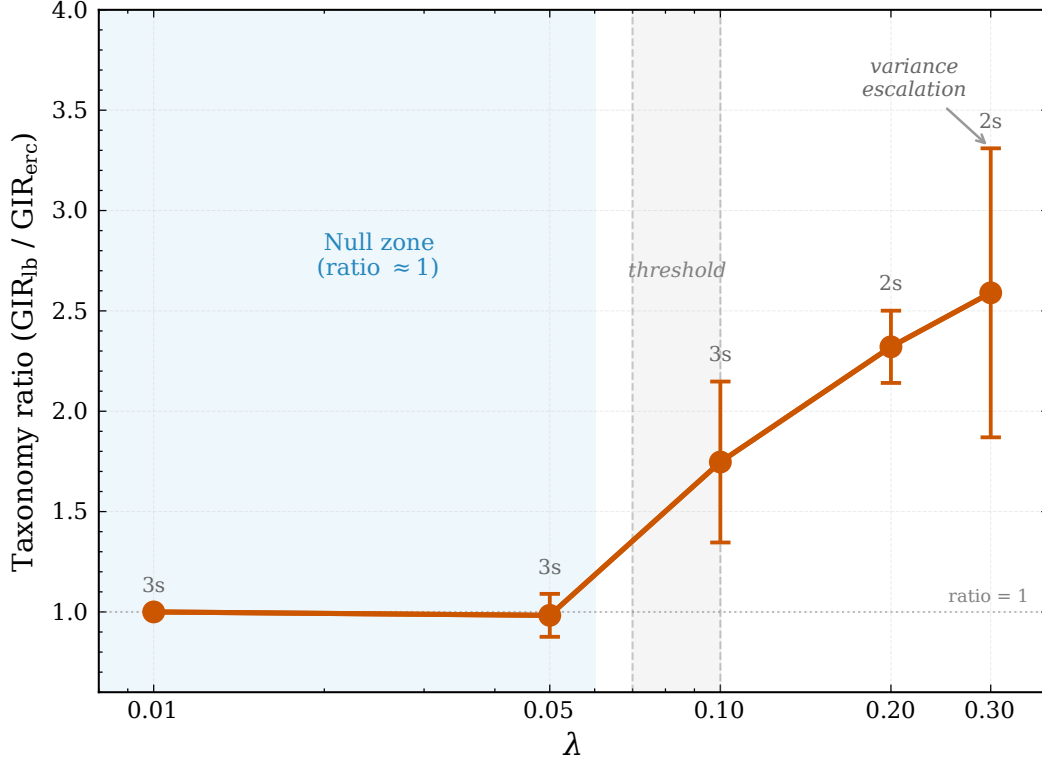


Figure 4: Dose-response curve of taxonomy ratio (mean GIR_{lb} / mean GIR_{erc}, unified terminal metric at epoch 30) vs. λ on Tiny-ImageNet. Below $\lambda=0.05$ (null zone), LB and ERA are indistinguishable; above $\lambda \approx 0.1$ (threshold), a monotonically increasing separation emerges with growing cross-seed variance at high λ . Point labels: seed count; error bars: ± 1 std where $n \geq 2$.

200 4.2.6 GIM-Switch Empirical Validation

201 We validate GIM-Switch (Section 3.4) on CIFAR-100 with elevated $\lambda_{lb}=0.1$ ($\tau_1=0.1$, $\tau_2=1.0$,
 202 $\gamma=0.5$, 90 epochs). GIM-Switch executes four tier transitions (seed = 42), including a bidirectional
 203 Monitor→Safe de-escalation at epoch 25 that distinguishes it from unidirectional decay schedules.
 204 At standard $\lambda_{lb}=0.01$, eGIR stays in $[0.003, 0.022]$ —deep in Safe—confirming no intervention is
 205 needed. Cross-seed replication ($n=2$) shows qualitatively consistent U-shaped eGIR trajectories.

206 4.2.7 Phantom GIM: Counterfactual Interference Under Loss-Free Balancing

207 Phantom GIM measures counterfactual LB interference on a model trained *without* LB updates
 208 (loss-free mode [Wang et al., 2024, DeepSeek-AI, 2024]; $n=2$ seeds). Phantom GIR reaches 3.3--
 209 $3.8\times$ in early training—*higher* than the matched actual GIR ($1.8\text{--}2.2\times$)—then drops below 1.0 by
 210 epoch 20–25 (Figure 13, Appendix C.17), confirming that joint training with LB actively reduces
 211 gradient conflict.

212 5 Discussion

213 5.1 Large GIR, Small Impact: The Attenuation Story

214 The headline numbers—GIR reaching $75\times$ —sound alarming, but the tax is far smaller than it appears.
 215 All auxiliary losses exhibit convergent GIR trajectories, with differential convergence rates forming
 216 a λ -dependent spectrum: load-balancing is slow-converging (terminal GIR ≈ 0.5) while ERA is
 217 fast-converging (terminal GIR < 0.1 within 30 epochs), yielding a $\sim 4\text{--}12\times$ mid-training separation
 218 at $\lambda \geq 0.1$. Effective GIR reveals this spectrum by separating λ -attenuated contributions from the
 219 denominator collapse that inflates raw GIR to $16\text{--}75\times$ post-saturation (Appendix J). During active

learning, eGIR stays below $0.02\times$ post-warmup; the maximum ($0.75\times$) occurs only at epoch 90 when accuracy is 99.97%—a post-saturation artifact, not evidence of gradient interference. Accuracy differences coexist with $\text{eGIR} < 1$: the pattern is consistent with routing policy reshaping rather than gradient-level competition. The U-shaped late-phase GIR trajectory is denominator-driven (task gradient drops $\sim 130\times$; auxiliary changes by $\sim 1.5\times$), specific to slow-converging losses—mid-training comparisons remain valid (Appendix J).

5.2 When Should Practitioners Worry?

The tax is bounded at micro scale—but benignity conditions are **not scale-invariant**: larger scales could shift eGIR above $1.0\times$ during active learning [Fedus et al., 2022, Zoph et al., 2022]. The convergence-rate taxonomy motivates a three-tier rule: **Safe** ($\text{eGIR} < 0.1$), **Monitor** ($0.1 \leq \text{eGIR} < 1.0$), **Intervene** ($\text{eGIR} \geq 1.0$), validated by GIM-Switch’s bidirectional tier transitions (Section 4.2.6). All ViT-Tiny conditions fall in Safe/Monitor; the taxonomy ordering is confirmed at ~ 98 M (Section 4.2.5) but is λ -threshold mediated. Both ViT and ConvNeXt-MoE achieve stabilizer-tier improvement at $\lambda=0.2$ (+1.47 pp and +1.95 pp), but regime boundaries are architecture-dependent (Appendix E): practitioners should not assume λ optima transfer across backbone families.

5.3 Limitations

Scale and statistical power. Models range from ≈ 7 M to ≈ 98 M, far from production MoE [Fedus et al., 2022, Zoph et al., 2022]. With $n=3-6$, Dunnett-corrected TOST establishes equivalence for all CIFAR-100 losses ($p \leq 0.045$); 2/6 ImageNet-50 comparisons remain inconclusive ($p \approx 0.051$); ImageNet-50 z-loss is ρ -sensitive (Appendix L).

Scope. Three auxiliary losses with AdamW only; Adam’s second-moment normalization may provide implicit attenuation (Appendix G).

6 Conclusion

We introduce GIM to decompose auxiliary loss gradient dynamics in sparse MoE, revealing a λ -dependent convergence spectrum: load-balancing converges slowly (terminal $\text{GIR} \approx 0.5$) while ERA self-extinguishes (terminal < 0.1), validated across three datasets and scales from ~ 7 M to ~ 98 M. This taxonomy is λ -threshold mediated, emerging above $\lambda \approx 0.1$ with monotonically increasing separation (13/13 sign test, $p < 0.001$). eGIR bounds the effective tax below $1\times$ during active learning; GIM-Switch operationalizes this with autonomous bidirectional tier transitions; phantom GIM quantifies the interference ($3.3-3.8\times$) that loss-free balancing sidesteps. Dunnett-corrected TOST bounds all CIFAR-100 losses within ± 2 pp ($p_{\text{Dunn}} \leq 0.045$); 2/6 ImageNet-50 comparisons remain inconclusive. All conclusions rest on $n=2-6$ seeds; larger-scale replication is needed.

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A GIM-Switch Technical Details

Ratio instability. As the model converges, $\|\mathbf{g}_{\text{task}}^{(l)}\|_2 \rightarrow 0$ faster than auxiliary gradients, causing GIR to diverge—a mathematical artifact rather than genuine interference growth. We quantify this via the *cross-layer max-to-mean ratio* of eGIR:

$$\rho(t) = \frac{\max_l \text{eGIR}^{\text{base}}(l, t)}{\frac{1}{L} \sum_{l=1}^L \text{eGIR}^{\text{base}}(l, t)}, \quad (8)$$

where L is the number of MoE layers. High ρ signals that a single layer dominates, typically due to denominator collapse rather than a true interference spike.

Smoothed eGIR signal. GIM-Switch uses the smoothed cross-layer maximum of base- λ eGIR:

$$\overline{\text{eGIR}}(t) = \frac{1}{W} \sum_{w=0}^{W-1} \max_l \text{eGIR}^{\text{base}}(l, t-w), \quad (9)$$

which attenuates transient spikes: in our ViT-Tiny experiments, raw eGIR of 1.587 at epoch 75 was smoothed to 0.814, preventing a premature tier transition.

Base- λ eGIR under adaptive scheduling. When λ_i is fixed, $\text{eGIR} = \lambda_i \cdot \text{GIR}_i$ is unambiguous. Under adaptive scheduling, λ_i varies at runtime. We define eGIR using the *initial* coefficient $\lambda_i^{(0)}$:

$$\text{eGIR}_i^{\text{base}}(l, t) = \lambda_i^{(0)} \cdot \text{GIR}_i(l, t), \quad (10)$$

providing a *counterfactual* diagnostic: the interference that *would* result if the original λ were restored.

B Method Details

B.1 Gradient Interference Variance (GIV)

To test whether multi-loss co-existence amplifies gradient variance, we define the **Gradient Interference Variance (GIV)** ratio. Let $\mathbf{g}_{\text{comb}}^{(l)} = \sum_{j=1}^K \lambda_j \mathbf{g}_j^{(l)}$ denote the combined auxiliary gradient over B mini-batches:

$$\text{GIV}_{\text{ratio}}(l) = \frac{\text{Tr}\left(\text{Cov}_B\left(\mathbf{g}_{\text{comb}}^{(l)}\right)\right)}{\sum_{j=1}^K \text{Tr}\left(\text{Cov}_B\left(\lambda_j \mathbf{g}_j^{(l)}\right)\right)}. \quad (11)$$

A ratio ≈ 1 indicates independent variances; values > 1 indicate amplification from loss interaction. In our experiments, the ratio remains within $[0.92, 1.06]$ (Appendix C.3), confirming that interference operates through magnitude dominance rather than stochastic amplification.

B.2 GIV: Detailed Interpretation

The GIV ratio (Eq. 11) tests whether co-existing auxiliary losses create *super-additive* gradient variance through cross-loss covariance terms. The variance of the combined gradient decomposes as:

$$\text{Tr}\left(\text{Cov}_B\left(\mathbf{g}_{\text{comb}}^{(l)}\right)\right) = \sum_{j=1}^K \text{Tr}\left(\text{Cov}_B\left(\lambda_j \mathbf{g}_j^{(l)}\right)\right) + 2 \sum_{j < k} \text{Tr}\left(\text{Cov}_B\left(\lambda_j \mathbf{g}_j^{(l)}, \lambda_k \mathbf{g}_k^{(l)}\right)\right).$$

When cross-covariance terms vanish, $\text{GIV}_{\text{ratio}} = 1$. Values exceeding 1 indicate positive cross-loss covariance (losses amplify each other’s variance); values below 1 indicate partial cancellation. In our experiments, the GIV ratio remains within $[0.92, 1.06]$ throughout training, indicating that cross-covariance terms are negligible and gradient interference operates through systematic magnitude effects rather than stochastic amplification.

B.3 Architecture Details

Input images of size 32×32 are tokenized into 4×4 patches, yielding 64 patch tokens augmented with a learnable [CLS] token and additive positional embeddings. A linear classification head maps the final [CLS] representation to class logits. Top-2 expert indices are selected per token and their outputs are combined via normalized gating weights.

B.4 GIM Measurement Protocol

At each measurement point, we perform a single-batch forward pass, followed by separate backward passes for each loss component with gradient accumulation disabled between passes. Router parameter gradients are extracted per layer and concatenated into a single vector per loss, from which we compute the pairwise cosine similarities (Eq. 3), gradient norms, and GIR values (Eq. 4).

GIV is estimated by repeating this procedure over 16 mini-batches and computing the covariance trace ratio (Eq. 11). For each mini-batch, per-loss gradient vectors are computed at each MoE layer, and the covariance matrices of individual and combined gradients are estimated across the 16-sample pool.

A limitation of single-batch GIR and AA estimation is that angular measurements become noisy when gradient norms approach zero. In late training, the near-vanishing task gradient amplifies AA measurement variance, which we account for in our analysis (Section 4.1.1).

B.5 GCF Interpretation

The Gradient Contribution Fraction (Eq. 5) extends GIR by incorporating the loss coefficient λ_i , measuring each component’s effective share of the combined router gradient rather than its raw magnitude ratio. While GIR captures whether an auxiliary gradient is *large* relative to the task gradient, GCF captures whether it is *dominant* in the actual parameter update. A loss with high raw GIR but small λ may have negligible GCF, explaining why GIR escalation does not necessarily translate to accuracy impact. The task fraction f_{task} serves as the reference: at epoch 90 with $\lambda_{\text{era}}=0.1$, the task gradient retains $\sim 72\%$ of the router update despite GIR_{lb} reaching $75\times$ (Figure 9).

B.6 Directional Effective GIR (deGIR)

While eGIR (Eq. 6) captures λ -weighted magnitude, it does not account for directional relevance. The **directional effective GIR (deGIR)** incorporates angular alignment:

$$\text{deGIR}_i(l, t) = \lambda_i \cdot \text{GIR}_i(l, t) \cdot |\cos \theta_i^{(l)}|, \quad (12)$$

where $\cos \theta_i^{(l)} = \text{GIM}^{(l)}[\text{task}, i]$; the absolute value measures interference magnitude regardless of gradient direction. When $\text{deGIR} \ll 1$, the auxiliary loss contributes negligibly to the task-relevant router update even if raw GIR is large. deGIR disambiguates regimes where high eGIR reflects real interference (amplifier, $|\cos \theta| > 0$) from directionally inert inflation (ghost-gradient, $|\cos \theta| \approx 0$; Section 4.1.2).

C CIFAR-100 Supplementary Analysis

C.1 GIR U-Shape Per-Epoch Details

The U-shaped GIR_{lb} trajectory (Figure 2) spans three phases across six seeds: $0.6\text{--}3.0\times$ at epoch 10, dropping to $0.3\text{--}1.1\times$ by epoch 30, then rising to $16\text{--}75\times$ (median $52.0\times$) by epoch 90. The escalation is consistent with asymmetric gradient decay: the task gradient collapses $> 100\times$ while the LB gradient decays only $5\text{--}12\times$. One outlier seed shows router collapse ($\text{GIR}_{\text{lb}}=2.8\times$ at epoch 90, deviating from the $16\text{--}75\times$ range of the remaining five seeds; see Section C.5). After λ -weighting ($\lambda_{\text{lb}}=0.01$), the effective LB contribution reaches $0.50 \pm 0.22\times$ at epoch 90—below unity, confirming that λ -attenuation bounds the effective interference despite raw GIR escalation.

C.2 Seed Verification of Angular Alignment

The angular alignment $AA(lb, era)$ is consistently negative across all λ values, with a broad trend toward weaker conflict at higher λ_{era} ($AA = -0.30$ at $\lambda_{era} = 0.025$ vs. -0.09 at $\lambda_{era} = 0.5$). However, the relationship is *not* monotonic: $\lambda_{era} = 0.1$ yields $AA = -0.22$, more negative than the adjacent $\lambda_{era} = 0.05$ at -0.19 . Cross-seed verification (seed 123) indicates that while the sign of AA is reproducible, the magnitude is seed-sensitive. Seed 123 at three representative λ values yields $AA(lb, era)$ of -0.16 ($\lambda = 0.025$), -0.18 ($\lambda = 0.1$), and -0.08 ($\lambda = 0.5$) at Layer 2, epoch 30—preserving the negative sign but differing in magnitude by up to 0.14 from seed 42. This non-monotonicity and seed dependence support the finding that AA is an unreliable diagnostic for auxiliary loss interference (Section 4.1.1).

Seed verification across three λ values with seed 123 further supports the finding that the GIR patterns—U-shape trajectory and low- λ amplification—are reproducible across seeds, supporting their interpretation as systematic phenomena rather than stochastic artifacts.

C.3 Gradient Variance Amplification (GIV Ratio)

We measured gradient variance amplification ($GIV \text{ ratio} = \text{Tr}(\text{Cov}(\mathbf{g}_{\text{combined}})) / \sum_i \text{Tr}(\text{Cov}(\mathbf{g}_i))$) to test whether the multi-loss gradient introduces multiplicative destabilization beyond additive interference. The GIV ratio remained within $[0.92, 1.06]$ throughout training across all configurations, indicating no evidence of variance amplification. A ratio near 1.0 implies that the combined gradient’s variance is well-approximated by the sum of individual variances, consistent with near-orthogonal gradient directions in late training.

C.4 Low- λ Amplification

A counter-intuitive finding emerges when comparing extended training at $\lambda_{era} = 0.025$ versus $\lambda_{era} = 0.1$ over 90 epochs. Despite a $4\times$ reduction in the ERA coefficient, GIR_{lb} at Layer 2 reaches 109.87 at $\lambda_{era} = 0.025$ compared to 34.96 at $\lambda_{era} = 0.1$ —a $3\times$ amplification of LB dominance (Table 3). At Layer 4, the pattern holds: GIR_{lb} reaches 32.47 versus 15.00.

Table 3: Extended 90-epoch GIM diagnostics comparing $\lambda_{era} = 0.025$ and $\lambda_{era} = 0.1$. GIR_{lb} is measured at the final epoch. Task gradient norm is measured on router parameters at Layer 2.

λ_{era}	GIR_{lb} (L2)	GIR_{lb} (L4)	Task Grad Norm (L2)	Test Acc
0.025	109.87	32.47	0.000627	0.452
0.1	34.96	15.00	0.0035	0.455

The mechanism operates through the denominator of the GIR ratio rather than the numerator. At low λ_{era} , the weaker ERA regularization imposes less constraint on routing parameters, allowing the model to overfit the task more freely—the task gradient norm at epoch 90 collapses to 0.000627, compared to 0.0035 at $\lambda_{era} = 0.1$ (a $5.6\times$ difference). The raw ERA gradient norm at epoch 10 is *larger* at low λ (0.081 at $\lambda_{era} = 0.025$ vs. 0.036 at $\lambda_{era} = 0.1$), confirming that reduced λ does not weaken the gradient itself but rather the regularization effect. Meanwhile, the LB gradient norm at epoch 90 is modestly smaller at low λ (0.069 vs. 0.121), but the $5.6\times$ faster collapse of the task gradient overwhelms this reduction, producing a $3\times$ higher GIR.

Test accuracy is virtually identical between the two settings (0.452 vs. 0.455), suggesting that the GIR explosion occurs after the critical learning window. This complicates the assumption that lower auxiliary loss coefficients are uniformly safer: while they reduce the direct auxiliary gradient magnitude, they can paradoxically amplify the *relative* dominance of other auxiliary losses.

C.5 Extended AA Non-Persistence Analysis

The angular alignment $AA(lb, era)$ between the two auxiliary gradients oscillates near zero throughout the 90-epoch trajectory, with no sustained directional trend. By the final training phase, $AA(lb, era)$ fluctuates within a narrow band, confirming that the directional conflict observed at epoch 30 in the dose-response analysis (Section 4.1) is transient rather than systematic. Similarly, $AA(lb, task)$ is

highly volatile in late training (± 0.41), reflecting the near-zero task gradient magnifying angular measurement noise rather than meaningful directional conflict.

The AA oscillation pattern is consistent across seeds: seed 42 and seed 123 both show near-zero mean AA(lb,era) in the final 30 epochs, with comparable variance. This reproducibility supports the interpretation that the lack of persistent directional alignment is a structural property of the differential convergence pattern—once the ERA gradient has largely self-extinguished, its residual direction is effectively random relative to the LB gradient.

Differential convergence dynamics: gradient decay rates. ERA follows a convergent gradient path, with gradient norm decaying from 0.036 to 0.001 over 90 epochs as the router-expert alignment objective is progressively satisfied. LB follows a persistent path (gradient norm from 0.6 to 0.05–0.12), never fully self-extinguishing because the uniform utilization objective is structurally unattainable. AA(lb,era) oscillates near zero throughout training, confirming that late-training interference is magnitude-driven rather than directional. Reducing λ_{era} counter-intuitively amplifies GIR_{lb} : at $\lambda=0.025$, GIR_{lb} reaches $109\times$ compared to $35\times$ at $\lambda=0.1$ (Table 3), associated with accelerated task gradient collapse—weaker ERA regularization allows the model to overfit faster, collapsing the task gradient denominator.

C.6 Cross-Architecture Probe: ConvNeXt-MoE

As a cross-architecture comparison, we repeat the four-condition experiment on a ConvNeXt-MoE backbone [Liu et al., 2022] (10.59M parameters, batch normalization, CNN-native architecture with MoE layers replacing selected depthwise convolution stages). Table 4 reports per-seed results.

Table 4: ConvNeXt-MoE on CIFAR-100 (best validation accuracy, 90 epochs, $n=3$ seeds). Mean $\pm$ std across seeds; $\bar{\Delta}$: seed-averaged difference vs. no-aux baseline. eGIR from seed 42 only.

Condition	s42	s123	s256	Mean $\pm$ Std (%)	$\bar{\Delta}$ (pp)	eGIR (s42)
No-Aux	56.76	57.76	58.73	57.75 ± 0.99	—	—
ERA ($\lambda=0.1$)	58.76	59.89	61.15	59.93 ± 1.20	+2.18	—
LB ($\lambda=0.01$)	56.09	56.66	58.22	56.99 ± 1.12	−0.76	$1.17\times$
Z-loss ($\lambda=0.001$)	56.99	57.65	58.65	57.76 ± 0.83	+0.01	—

ERA consistently positive across seeds. ERA improves accuracy over the no-aux baseline in all three seeds: +2.00 pp (s42), +2.13 pp (s123), +2.42 pp (s256), with a mean $\bar{\Delta}_{\text{ERA}} = +2.18$ pp. This effect is larger than the matched ViT-Tiny ERA result at $\lambda=0.1$ ($\bar{\Delta} = +0.76$ pp, $n=3$; Table 9; the $n=6$ ViT-Tiny mean of +0.25 pp in Table 16 includes additional seeds with negative Δ , making the n -matched comparison more appropriate). One possible explanation is the architectural difference (batch normalization vs. layer normalization, or the different MoE layer positions); however, the models also differ in parameter count (10.59M vs. 6.86M), so the source of the stronger ERA benefit cannot be isolated from these data alone.

LB reaches Intervene tier in 1 of 3 seeds. At seed 42, LB produces an effective GIR of $1.17\times$ —exceeding $1.0\times$ and placing it in the **Intervene** tier of the three-tier heuristic (Section 5.2). The remaining seeds are near-threshold ($0.94\times$ at s256) or below ($0.56\times$ at s123), indicating the Intervene boundary is seed-sensitive. All three seeds show negative Δ : −0.67 pp (s42), −1.10 pp (s123), −0.51 pp (s256), mean $\bar{\Delta}_{\text{LB}} = -0.76$ pp—accuracy degradation occurs even when eGIR < 1 , suggesting the threshold is conservative.

Z-loss non-significant ($n=3$). Z-loss yields +0.23 pp (s42), −0.11 pp (s123), −0.08 pp (s256), with a mean $\bar{\Delta} = +0.01$ pp—consistent with the ViT-Tiny finding (ViT-Tiny $\bar{\Delta} = +0.69$ pp, $p=0.205$) that high raw GIR with near-zero angular alignment does not translate to an accuracy cost (Section 4.1.2). Gradient magnitude alone does not imply interference when the auxiliary and task gradients are approximately orthogonal.

509 C.7 ConvNeXt-MoE ERA λ Dose-Response

510 To characterize how the ERA coupling coefficient governs accuracy and gradient dynamics on a non-
 511 Transformer backbone, we sweep $\lambda_{\text{era}} \in \{0, 0.001, 0.01, 0.05, 0.1, 0.2, 0.5, 1.0\}$ on ConvNeXt-MoE
 512 (CIFAR-100, $n=3$ seeds). We report ConvNeXt-MoE GIR at Layer 3 (L3) rather than L2 because the
 513 L2 task gradient is $\sim 47\times$ smaller than L3, inflating GIR through the denominator effect rather than
 514 reflecting genuine auxiliary interference (Mechanism B; Section 5.1). Table 5 and Figure 5 report
 515 results; Figure 6 juxtaposes the ViT-Tiny and ConvNeXt-MoE curves.

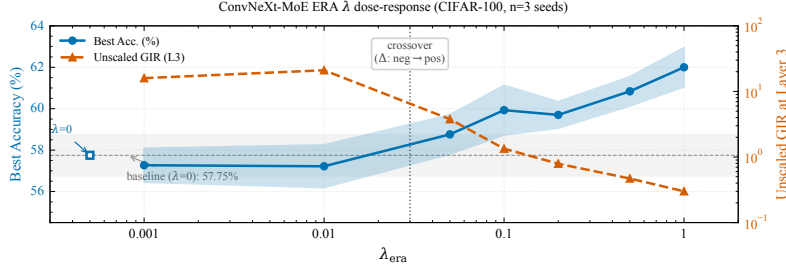


Figure 5: Accuracy and GIR as a function of ERA λ for ConvNeXt-MoE on CIFAR-100 ($n=3$ seeds, mean $\pm$ std). Left axis: best validation accuracy; right axis: unscaled GIR at Layer 3. Higher λ generally improves accuracy while GIR self-extinguishes.

Table 5: ConvNeXt-MoE ERA λ dose-response (CIFAR-100, $n=3$ seeds). Accuracy: mean $\pm$ std across seeds. GIR and cosine at Layer 3, epoch 90.

λ_{era}	Mean Acc (%)	$\bar{\Delta}$ (pp)	Std(Δ)	GIR (L3)	cos (L3)
0 (none)	57.75 $\pm$ 0.99	—	—	—	—
0.001	57.27 $\pm$ 0.82	−0.48	0.32	15.95 $\pm$ 9.49	0.02
0.01	57.22 $\pm$ 1.03	−0.53	0.77	21.05 $\pm$ 12.74	0.00
0.05	58.76 $\pm$ 0.95	+1.01	0.11	3.78 $\pm$ 3.30	−0.02
0.1	59.93 $\pm$ 1.20	+2.18	0.22	1.34 $\pm$ 0.28	0.00
0.2	59.70 $\pm$ 0.64	+1.95	0.35	0.79 $\pm$ 0.16	−0.01
0.5	60.84 $\pm$ 0.71	+3.09	0.92	0.47 $\pm$ 0.05	0.00
1.0	62.00 $\pm$ 0.95	+4.25	0.09	0.30 $\pm$ 0.12	−0.03

516 **Dose-response trend.** Three-seed mean accuracy increases broadly from $\lambda=0.01$ to $\lambda=1.0$, span-
 517 ning a 4.78 pp range ($\bar{\Delta}$ from -0.53 to $+4.25$ pp) with local fluctuations within seed uncertainty
 518 ($\lambda=0.2$ dips slightly below $\lambda=0.1$). At low λ (0.001, 0.01), the coupling signal is insufficient to
 519 improve routing quality, yielding neutral-to-negative effects; at $\lambda \geq 0.05$, the benefit exceeds seed-
 520 level noise. The crossover from negative to positive Δ occurs between $\lambda=0.01$ and $\lambda=0.05$. The
 521 gain continues through high λ : $+3.09$ pp at $\lambda=0.5$ and $+4.25$ pp at $\lambda=1.0$, a $+1.16$ pp increment
 522 suggesting that stronger coupling continues to benefit ConvNeXt-MoE even in the saturation regime.

523 **Self-limiting gradient dynamics.** GIR broadly decreases from $15.95\times$ at $\lambda=0.001$ through $21.05\times$
 524 at $\lambda=0.01$ to $0.30\times$ at $\lambda=1.0$ (L3, $n=3$ mean)—a $\sim 53\times$ reduction across the effective range ($\lambda \geq$
 525 0.01)—confirming the self-limiting property at the cross-architecture level: higher λ is associated
 526 with faster ERA convergence, which extinguishes the gradient signal. (The L3 GIR at $\lambda=0.01$
 527 slightly exceeds $\lambda=0.001$, but the overall decreasing trend from low to high λ is clear.) At $\lambda=1.0$,
 528 GIR falls below $1\times$ ($0.30\times$, L3), meaning the ERA gradient is smaller than the task gradient
 529 even in the post-saturation regime. The extreme L2 GIR at $\lambda=0.001$ ($646\times$ at L2; $15.95\times$ at L3)
 530 reflects the noise regime where the coupling constraint is unsatisfied rather than meaningful gradient
 531 interference (cosine ≈ 0). The ERA loss itself exhibits strict positive correlation between λ and
 532 convergence degree: final ERA loss ranges from ≈ -0.365 at $\lambda=0.001$ (barely converged) through
 533 -0.42 ($\lambda=0.01$) and -0.62 ($\lambda=0.05$) to ≈ -0.995 at $\lambda=0.5$ (near-complete convergence). This
 534 reveals an empirical chain: $\lambda \uparrow \rightarrow$ constraint strength $\uparrow \rightarrow$ ERA converges $\rightarrow$ gradient vanishes

→ routing quality improved as byproduct. At $\lambda=0.001$, L3 GIR is high ($15.95\times$) but cosine alignment is near-zero (0.02), producing an effectively orthogonal gradient whose accuracy effect is architecture-dependent (Section E)—the constraint is too weak for the network to satisfy, so the residual gradient is large but directionally random (noise). The L2 GIR at this λ reaches $646\times$ due to BN-induced task-gradient attenuation (Mechanism B), but the L3 value reflects the operationally relevant layer. At $\lambda=0.5$, L3 GIR drops to $0.47\times$ with near-zero cosine (0.00), indicating that both gradient magnitude and direction are benign post-convergence—the routing improvement is already in place and the residual gradient is neither large nor directionally aligned. The effective magnitude ($\lambda \cdot \text{GIR}_{\text{L3}} = 0.5 \times 0.47 = 0.24\times$) confirms the parameter update is task-dominated; the routing benefit was established during convergence, not sustained by the residual gradient. At $\lambda=1.0$, L3 GIR drops further to $0.30\times$ with cosine -0.03 —at near-complete convergence (ERA loss ≈ -0.997), both magnitude and direction of the residual gradient approach zero. (L2 cosine at this regime shows strong anti-alignment (-0.41 to -0.88 across seeds), but this reflects the BN-attenuated layer where task gradients are small; see Appendix K for the L2 analysis.) This self-limiting property distinguishes convergent losses from persistent ones: load-balancing gradients do not self-extinguish with increasing λ because the uniformization constraint is never fully satisfied, so higher λ_{lb} monotonically increases gradient pressure without a compensating decay mechanism. For convergent losses, the “higher λ = higher risk” intuition inverts.

Caveats. The broadly increasing dose-response curve is confirmed across $n=3$ seeds on a single dataset. Multi-seed data show the increasing pattern holds in aggregate: mean $\bar{\Delta}$ rises from -0.48 pp ($\lambda=0.001$) to $+4.25$ pp ($\lambda=1.0$), though per-seed responses vary (Std(Δ) up to 0.92 pp at $\lambda=0.5$). Multi-seed data at $\lambda=0.2$ (Appendix C.8) shows the increasing pattern does not hold uniformly across seeds: comparing $\lambda=0.1$ (Table 4) to $\lambda=0.2$ (Table 6), s123 accuracy decreases by 0.19 pp and s256 by 0.81 pp, even though s42 continues to increase ($+0.31$ pp). The dose-response peak is likely architecture- and seed-dependent. The observed accuracy range (56.07 – 61.00%) exceeds the multi-seed variance of the $n=3$ ConvNeXt-MoE experiments (Table 4), suggesting the λ effect is real but seed-sensitive. Notably, final training accuracy is 99.97% across all λ values, ruling out reduced overfitting as an alternative explanation for the accuracy gains at high λ —the benefit is routing-mediated, not regularization-mediated.

C.8 Cross-Architecture Stabilizer Verification: ConvNeXt-MoE at $\lambda=0.2$

To verify whether the stabilizer regime identified on ViT-Tiny (Section 5.2) generalizes across architectures, we evaluate ERA at $\lambda_{\text{era}}=0.2$ on ConvNeXt-MoE with three seeds. Table 6 extends the main-text summary (Table 9) with per-seed gradient diagnostics.

Table 6: ConvNeXt-MoE stabilizer verification at ERA $\lambda=0.2$ (CIFAR-100, 3 seeds). Baseline accuracy from the no-aux condition (Table 4). GIR and cosine similarity measured at Layer 3, epoch 90.

Seed	Baseline (%)	ERA $\lambda=0.2$ (%)	Δ (pp)	GIR (L3)	cos (L3)
42	56.76	59.07	+2.31	0.89	-0.03
123	57.76	59.70	+1.94	0.87	-0.04
256	58.73	60.34	+1.61	0.60	+0.03
Mean	57.75 (std 0.99)	59.70 (std 0.64)	+1.95	—	—

Consistent positive effect and standard deviation reduction. All three seeds yield positive Δ , ranging from $+1.61$ pp (s256) to $+2.31$ pp (s42), with a mean of $+1.95$ pp. The stabilizer effect is confirmed: cross-seed standard deviation of absolute accuracy tightens from 0.99 pp (baseline) to 0.64 pp (ERA $\lambda=0.2$)—a 35% standard deviation reduction. Compared to ViT-Tiny at the same λ ($\bar{\Delta} = +1.47$ pp, Acc Std = 0.80), the ConvNeXt-MoE effect is both stronger ($+1.95$ vs. $+1.47$ pp) and tighter (Acc Std = 0.64 vs. 0.80), directionally consistent, providing evidence that the stabilizer regime extends to at least one non-Transformer architecture.

Gradient diagnostics. GIR at Layer 3 ranges from $0.60\times$ to $0.89\times$ across seeds—well below the ghost-gradient ($\sim 48\times$ on ViT, $\sim 16\times$ on ConvNeXt at L3) and amplifier regimes, confirming

operation within the stabilizer zone where GIR is moderate and the ERA gradient has partially self-extinguished. Cosine alignment at L3 is near-zero across seeds (+0.03 at s256, -0.03 at s42, -0.04 at s123) without producing corresponding accuracy divergence (Acc Std = 0.64 pp), reinforcing the finding that post-saturation angular alignment is a noisy diagnostic at this scale: the routing improvement has already been established during convergence, so the residual gradient direction is functionally inert.

Caveats. This verification uses $n=3$ seeds on a single dataset (CIFAR-100) at a single λ value (0.2). While the results are directionally consistent with the ViT-Tiny stabilizer regime, they do not establish a universal cross-architecture stabilizer property. The ConvNeXt-MoE and ViT-Tiny backbones differ in normalization (batch vs. layer), MoE layer position, and parameter count (10.59M vs. 6.86M), so the source of the stronger ConvNeXt-MoE effect cannot be isolated from these data alone.

C.9 Cross-Architecture λ Response Comparison

Figure 6 juxtaposes the $n=3$ seeds ERA λ dose-response curves for ViT-Tiny and ConvNeXt-MoE, highlighting a qualitative difference in λ sensitivity across architectures.

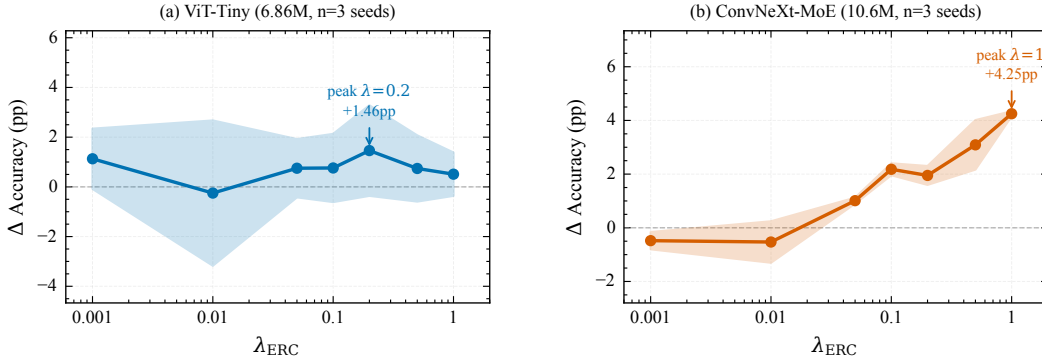


Figure 6: Cross-architecture ERA λ dose-response on CIFAR-100 ($n=3$ seeds, mean $\pm$ std). **(a)** ViT-Tiny exhibits an inverted-U response with peak at $\lambda=0.2$ (+1.46 pp); **(b)** ConvNeXt-MoE shows broadly increasing improvement up to $\lambda=1.0$ (+4.25 pp). The contrasting shapes suggest that optimal λ is architecture-dependent: Transformer-based routers saturate at moderate coupling, while CNN-based routers continue to benefit from stronger ERA constraints.

C.10 GIR and Cosine Alignment Across λ Values

Figure 7 shows the joint evolution of raw GIR and angular alignment (cosine similarity) across the ERA λ sweep for ViT-Tiny (seed 42, Layer 2, epoch 90) and ConvNeXt-MoE ($n=3$ mean, Layer 3, epoch 90). Both architectures exhibit inverse GIR- λ scaling (higher $\lambda \rightarrow$ lower GIR), but the cosine trajectories diverge qualitatively: ViT-Tiny shows oscillating negative alignment (reaching -0.59 at $\lambda=0.01$), while ConvNeXt-MoE L3 cosine remains near zero across all λ values ($|\cos \theta| \leq 0.03$). This contrast indicates that ConvNeXt-MoE ERA interference at the operationally relevant layer is magnitude-only—the auxiliary gradient is large at low λ but directionally random, producing no systematic directional conflict with the task gradient.

TOST sensitivity to equivalence margin. Table 7 reports TOST p -values across four equivalence margins $\delta \in \{1.0, 1.5, 2.0, 2.5\}$ pp for all six conditions. At our chosen $\delta=2$ pp, all six conditions pass at $\alpha=0.025$, consistent with the equivalence results reported in Table 1. At the tighter $\delta=1.5$ pp, only the two CIFAR-100 conditions (LB and ERA) pass at $\alpha=0.05$; none pass at $\alpha=0.025$. At $\delta=1.0$ pp, no condition passes. This pattern indicates that $\delta=2$ pp is the smallest margin at which equivalence is broadly demonstrable given our sample sizes ($n=5-6$), supporting its choice as a practically meaningful bound.

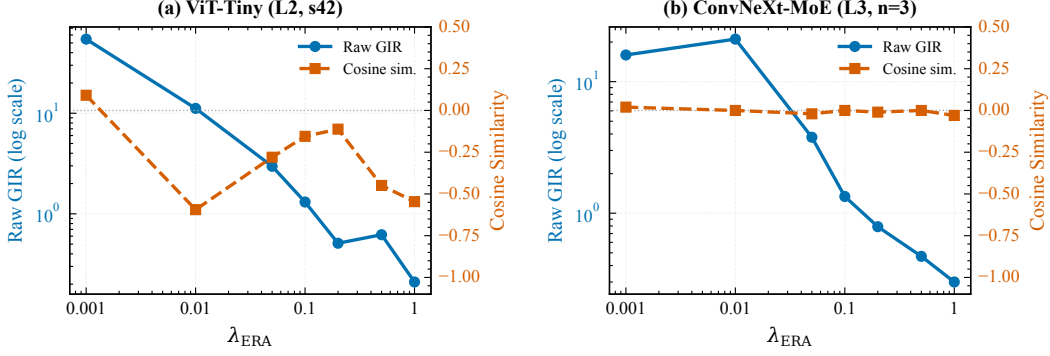


Figure 7: Raw GIR (left axis, log scale) and cosine similarity (right axis) between ERA and task gradients as a function of λ_{ERA} (epoch 90, CIFAR-100). **(a)** ViT-Tiny (Layer 2, seed 42): GIR drops from $54.6 \times$ ($\lambda=0.001$) to $0.21 \times$ ($\lambda=1.0$); cosine oscillates between $+0.09$ and -0.59 . **(b)** ConvNeXt-MoE (Layer 3, $n=3$ mean): GIR drops from $16.0 \times$ to $0.30 \times$; cosine remains near zero ($|\cos \theta| \leq 0.03$) across all λ values, indicating magnitude-only interference without directional conflict at the operationally relevant layer.

Table 7: TOST sensitivity to equivalence margin δ . Each cell shows the TOST p -value; $\checkmark$ denotes $p < 0.05$; $\checkmark\checkmark$ denotes $p < 0.025$ (one-sided TOST threshold). Accuracy metric matches Table 1: best checkpoint for CIFAR-100, final epoch for ImageNet-50. No multiple-comparison correction is applied; this is an exploratory sensitivity analysis.

Condition	$\delta=1.0$	$\delta=1.5$	$\delta=2.0$	$\delta=2.5$
CIFAR-100 LB ($n=6$)	.128	.037 $\checkmark$	.012 $\checkmark\checkmark$	.004 $\checkmark\checkmark$
CIFAR-100 ERA ($n=6$)	.107	.032 $\checkmark$	.011 $\checkmark\checkmark$	.004 $\checkmark\checkmark$
ImageNet-50 LB ($n=5$)	.185	.062	.023 $\checkmark\checkmark$	.010 $\checkmark\checkmark$
ImageNet-50 ERA ($n=5$)	.198	.065	.023 $\checkmark\checkmark$	.009 $\checkmark\checkmark$
CIFAR-100 Z-loss ($n=6$)	.269	.073	.020 $\checkmark\checkmark$	.006 $\checkmark\checkmark$
ImageNet-50 Z-loss ($n=6$)	.185	.062	.021 $\checkmark\checkmark$	.008 $\checkmark\checkmark$

607 C.11 Dose-Response Details

608 C.12 Gradient Contribution Fraction

609 When $\sum_j f_j > f_{\text{task}}$ (Eq. 5), auxiliary losses collectively dominate the router update.

610 C.13 Timing Mismatch Visualization

611 C.14 ERA λ Sensitivity Across Seeds (ViT-Tiny)

612 **Ghost gradient vs. amplifier: per-seed analysis.** At $\lambda=0.001$ ($n=3$), GIR_{era} at L2 explodes
613 post-saturation ($\sim 48\text{--}59 \times$ at epoch 90) with near-zero cosine alignment ($|\cos| < 0.10$), making the
614 ERA gradient effectively orthogonal to the task gradient—a large but directionally null residual. All
615 three seeds show positive Δ ($\bar{\Delta} = +1.13 \pm 0.54$ pp), confirming that the ghost-gradient regime does
616 not harm accuracy on ViT-Tiny. At $\lambda=0.01$ ($n=3$), GIR also escalates but with non-negligible cosine
617 alignment in the seed (s123) that collapses, producing seed-specific accuracy damage (amplifier
618 regime). The two λ values are indistinguishable on raw GIR alone: only when paired with angular
619 alignment does the diagnostic separate ghost-gradient (orthogonal noise) from amplifier interference.
620 Seed s123 yields negative Δ in 6 of 7 λ values, confirming that the ERA benefit is seed-dependent
621 rather than universal.

Table 8: Dose-response summary at epoch 30 for Layer 2. $\text{GIR}_{\text{era}} = \|\nabla \mathcal{L}_{\text{era}}\| / \|\nabla \mathcal{L}_{\text{task}}\|$. $\text{AA}(\text{lb}, \text{era})$ is the cosine similarity between LB and ERA gradients.

λ_{era}	ERA Grad Norm	GIR_{era}	$\text{AA}(\text{lb}, \text{era})$	Test Acc
0.025	0.053	0.21	−0.30	0.420
0.05	0.030	0.23	−0.19	0.426
0.1	0.017	0.11	−0.22	0.430
0.2	0.009	0.06	−0.14	0.430
0.3	0.007	0.04	−0.09	0.427
0.5	0.005	0.02	−0.09	0.422

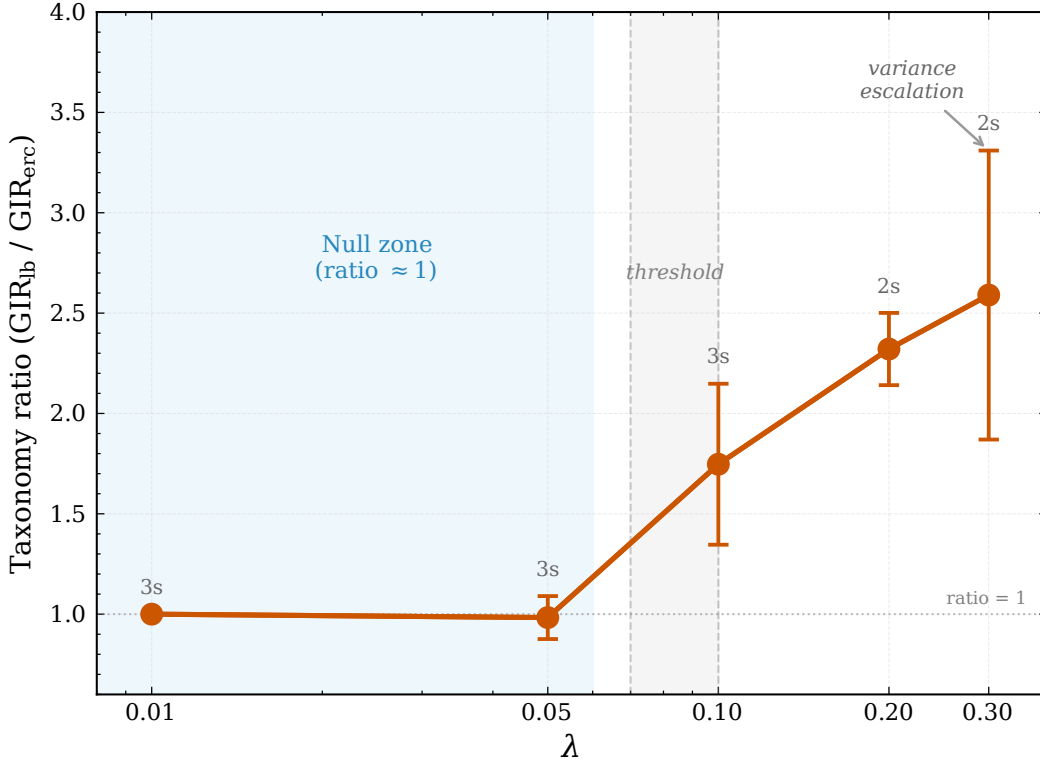


Figure 8: ERA gradient norm and GIR_{era} as a function of λ_{era} at epoch 30 (Layer 2). Higher coefficients accompany faster ERA convergence, self-extinguishing the gradient signal.

622 C.15 Raw vs. Effective GIR Across Loss Types

623 C.16 Z-Loss GIR Details (ViT-Tiny)

624 Table 10 reports per-seed z-loss GIR at epoch 90 across all $n=6$ seeds. Cross-seed GIR spans two
625 orders of magnitude ($2.8\text{--}375\times$ at L2), confirming that the variance is associated with seed-dependent
626 task gradient collapse rates rather than a single outlier.

627 **GIR trajectory.** Z-loss GIR trajectories (seeds 99 and 256, which have full per-epoch data) reveal
628 a non-monotonic pattern distinct from LB’s monotonic late-training explosion. At Layer 2, z-loss
629 GIR starts high (s99: $31.2\times$, s256: $50.6\times$ at epoch 1), drops to a mid-training trough ($2.6\text{--}5.5\times$ at
630 epoch 40), then rises again in late training ($24.0\text{--}127.0\times$ at epoch 90). This reflects z-loss’s persistent
631 lifecycle: the z-loss gradient does not self-extinguish (unlike ERA), but the late-training GIR rise
632 is associated with task gradient collapse rather than increasing z-loss gradient norms—the same
633 passive amplification mechanism as LB (Section 4.1.1). After λ -weighting ($\lambda_z=0.001$), the effective

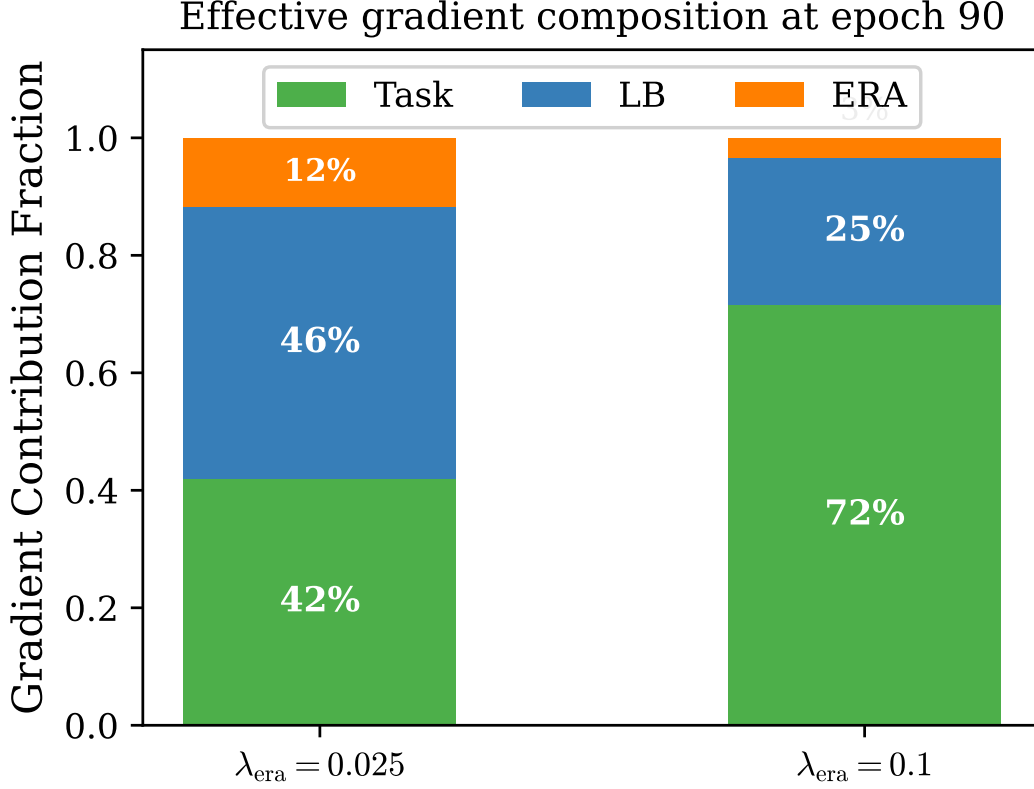


Figure 9: Gradient Contribution Fraction (GCF) stacked by loss component across training epochs (CIFAR-100 ViT-Tiny MoE, $\lambda_{\text{era}}=0.1$). As training progresses, the task gradient’s share of the effective router update diminishes while load-balancing increasingly dominates.

Table 9: ERA λ sensitivity: ViT-Tiny on CIFAR-100 (3 seeds, 90 epochs, best checkpoint). Baseline: $44.74 \pm 1.05\%$. The 3-seed baseline differs from the 6-seed estimate in Table 1 (45.3%) due to seed subset. ConvNeXt-MoE results in Table 4.

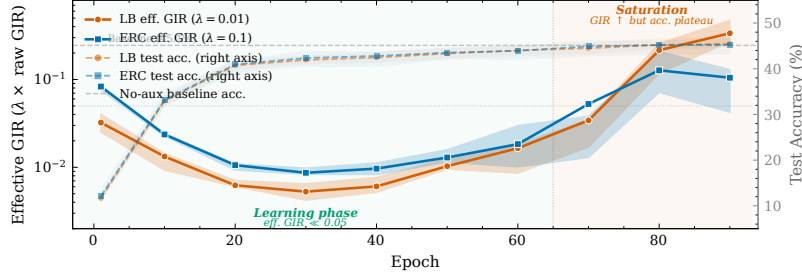
λ_{era}	Acc. (%)	$\bar{\Delta}$ (pp)
0.001	45.87 ± 0.68	+1.13
0.01	44.49 ± 1.96	−0.25
0.05	45.49 ± 0.75	+0.75
0.1	45.50 ± 0.36	+0.76
0.2	46.21 ± 0.80	+1.47
0.5	45.48 ± 0.66	+0.74
1.0	45.26 ± 0.56	+0.51

634 GIR remains $\leq 0.375\times$ across all seeds and layers (maximum: s1337 L2, $0.001 \times 375 = 0.375$),
635 approximately half of LB’s peak effective contribution ($\lambda_{\text{lb}} \cdot \text{GIR}_{\text{lb}} = 0.75\times$).

636 C.17 Phantom GIR Figure

637 D ImageNet-50 Detailed Analysis

638 This appendix provides detailed per-epoch trajectory analyses, directional alignment results, and
639 regime boundary discussion supporting the summary in Section 4.2.



Note: LB uses $\lambda = 0.01$, ERC uses $\lambda = 0.1$. Absolute effective GIR values are not directly comparable; focus on temporal trends. LB: 6 seeds; ERC: 5 seeds (s42 excluded — only 2/10 epochs recorded).

Figure 10: Effective GIR ($\lambda \times$ raw GIR) and test accuracy over 90 epochs on CIFAR-100 ViT-Tiny MoE. During the active learning phase (training accuracy $< 99\%$; shaded green, approximately epochs 5–50 for CIFAR-100), effective GIR remains below $0.02\times$ for both LB and ERA. GIR rises only during the saturation phase (red), when accuracy has already plateaued—the *timing mismatch* that explains why large raw GIR does not translate to accuracy degradation. LB: $n=6$ seeds, $\lambda_{lb}=0.01$; ERA: $n=5$ seeds, $\lambda_{era}=0.1$.

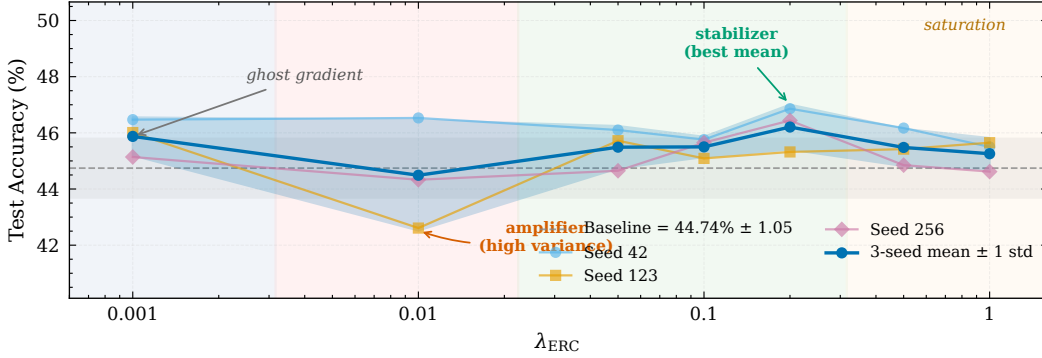


Figure 11: ViT-Tiny ERA λ sensitivity on CIFAR-100 across three random seeds (s42, s123, s256). 3-seed mean (line) $\pm$ std (band) and per-seed markers; gray band shows no-aux baseline mean (44.74%) $\pm$ std (1.05 pp). The optimum at $\lambda=0.2$ delivers mean 46.21% ($\Delta = +1.47$ pp) with cross-seed std (0.80 pp) below the baseline std—a stabilizer regime. Annotations mark behavioral regimes: ghost gradient ($\lambda=0.001$), amplifier ($\lambda=0.01$, high variance), stabilizer ($\lambda=0.2$, best mean), and saturation ($\lambda \geq 0.5$).

640 D.1 Per-Epoch Load-Balancing GIR Trajectories

641 Post-warmup, the LB GIR trajectories diverge sharply between regimes. On ImageNet-50, GIR_{lb}
642 generally declines but with local fluctuations: at Layer 2, from 2.06 at epoch 20 to a minimum
643 of 0.28 at epoch 70, then rising to 0.67 at epoch 80 before settling at 0.51 at epoch 90 (seed=42).
644 These fluctuations reflect stochastic variation in the task gradient norm rather than any systematic
645 trend—critically, GIR remains bounded at $\mathcal{O}(1)$ throughout training at both layers, with transient
646 excursions above $1\times$ (e.g., seed=123 Layer 2 at epoch 30, $GIR_{lb}=1.09$) but no sustained escalation.
647 Across both seeds, GIR_{lb} settles in the range $0.33\text{--}0.51\times$ at Layer 2 and $0.17\text{--}0.46\times$ at Layer 4 at
648 epoch 90. On CIFAR-100, GIR_{lb} follows a qualitatively different, non-monotonic trajectory: after an
649 early trough (0.57 at epoch 30), it rises through 1.93 at epoch 50 to 7.04 at epoch 60, *dips* to 1.23
650 at epoch 70 as the task gradient transiently recovers, then explodes to 23.4 at epoch 80 and 75.2 at
651 epoch 90 as sustained task gradient collapse sets in.

652 The gradient norms tell the full story. On ImageNet-50 L2, the LB gradient norm decays from 0.431 to
653 0.087 ; on CIFAR-100 L2, from 0.118 to 0.050 . While the absolute magnitudes differ ($\sim 3.7\times$ at epoch
654 1, reflecting different dataset complexities at 32×32), both exhibit the same qualitative behavior: the
655 LB gradient decays modestly but never vanishes. The GIR explosion in the high-overfitting regime is
656 consistent with *passive denominator collapse*—the task gradient collapsing by $\sim 150\times$ (from 0.107
657 to ~ 0.0007) while the LB gradient decays by only $\sim 2.4\times$ (from 0.118 to 0.050).

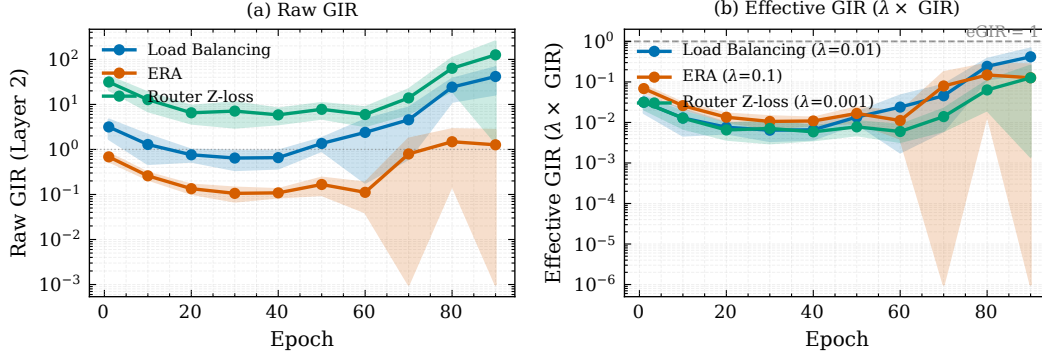


Figure 12: Raw GIR vs. effective GIR ($\lambda \times \text{GIR}$) at Layer 2 over 90 epochs (CIFAR-100 ViT-Tiny MoE). λ -weighting compresses all three loss types below $\text{eGIR} = 1$ throughout training, despite raw GIR reaching $75\times$ (LB) and $375\times$ (z-loss).

Table 10: Per-seed z-loss GIR at epoch 90 (ViT-Tiny, CIFAR-100, $\lambda_z=0.001$). Raw GIR before λ -weighting.

Seed	GIR _z (L2)	GIR _z (L4)
42	61.32	63.26
123	2.83	10.87
7	167.09	121.06
99	24.05	105.37
1337	374.98	207.99
256	126.99	84.43
Range	2.8–375.0 $\times$	10.9–208.0 $\times$

658 D.2 Directional Alignment Under Bounded GIR

659 The angular alignment $\text{AA}(\text{task}, \text{lb})$ at Layer 2, epoch 90 is -0.478 (seed=42) / -0.04 (seed=123)
660 on ImageNet-50—the seed-42 value indicating substantial directional conflict, the seed-123 value
661 near-orthogonal. Yet with GIR_{lb} at 0.51 and 0.33 respectively, the LB gradient’s magnitude is at most
662 half that of the task gradient in both seeds. This illustrates a key diagnostic principle from the GIM
663 framework (Section 3.1): directional conflict is operationally irrelevant when the conflicting gradient
664 is magnitude-subordinate. Angular alignment alone would flag seed 42 as a potential interference site
665 while discharging seed 123; GIR contextualizes both consistently as benign.

666 D.3 Per-Epoch ERA Trajectories

667 On ImageNet-50, GIR_{era} at Layer 2 decays from 1.22 at epoch 1 to 0.037 at epoch 90—a $33\times$
668 reduction. At Layer 4, the decay is even faster: from 1.66 to 0.024. On CIFAR-100, GIR_{era} at Layer
669 2 follows a broadly similar trajectory: from 0.55 at epoch 1 to 0.048 at epoch 90 (seed=42) or 0.355
670 (seed=123). The decay is not strictly monotonic—at epoch 80, the CIFAR-100 ERA GIR spikes from
671 0.038 at epoch 70 to 0.50 (seed=42) or 1.06 (seed=123), coinciding with a transient task gradient
672 collapse ($\|\nabla \mathcal{L}_{\text{task}}\|_2 = 0.004\text{--}0.005$). The seed=123 spike briefly exceeds $1\times$, demonstrating that
673 ERA’s bounded behavior is not absolute: under sufficiently severe task gradient collapse, even a
674 convergent loss can transiently dominate. However, this excursion is brief and modest—the maximum
675 GIR_{era} of $1.06\times$ is dwarfed by LB’s $23\times$ at the same epoch—confirming that the convergent lifecycle
676 fundamentally constrains ERA’s capacity for sustained dominance.

677 The contrast generalizes to a design principle. LB optimizes for a global distributional property
678 (uniform expert utilization) that is never fully achievable, leaving a persistent residual gradient. ERA
679 optimizes for a local alignment condition that can be approximately satisfied, after which the gradient
680 vanishes. This **convergent vs. persistent lifecycle** distinction (Section 4.1.1) determines whether an

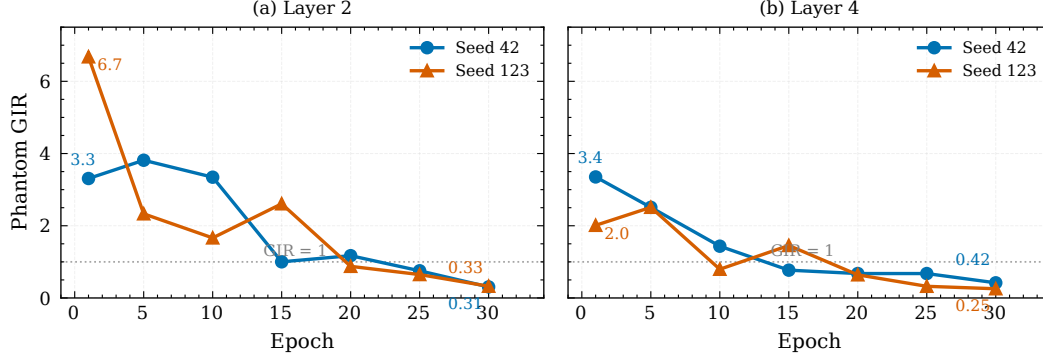


Figure 13: Phantom GIR under loss-free balancing ($n=2$ seeds). GIR declines from $3\text{--}7\times$ in early training to $< 0.5\times$ by epoch 30, confirming intrinsic convergence without LB gradient updates.

681 auxiliary loss is susceptible to passive GIR amplification: only losses with persistent gradients can
 682 become dominant when the task gradient collapses.

683 D.4 Regime Boundaries Are Dynamic

684 The 2×2 ablation indicates that the accuracy cost is regime-dependent, but the regime boundary
 685 itself is not a static property of the dataset or model. It is determined dynamically by the task gradient
 686 trajectory—specifically, whether $\|\nabla \mathcal{L}_{\text{task}}\|$ undergoes the sustained collapse that enables passive GIR
 687 amplification.

688 Several implications follow. First, the same architecture trained on the same dataset could transition
 689 from the low-overfitting to the high-overfitting regime as training progresses—for example, a ViT-
 690 Tiny on ImageNet-50 trained for 300+ epochs as the model begins to memorize. The GIR trajectory
 691 serves as a *real-time regime indicator*: a rising post-warmup GIR_{lb} trend signals the onset of task
 692 gradient collapse and the approaching dominance of the accuracy cost.

693 Second, the regime classification is not synonymous with dataset scale. ImageNet-50 with a $\sim 40\text{pp}$
 694 generalization gap is far from an idealized “well-fit” setting, yet its task gradient dynamics are
 695 sufficient to suppress LB gradient interference. The operative distinction is whether the task loss
 696 landscape retains curvature at the router parameters—a condition that depends on model capacity,
 697 data complexity, augmentation strategy, and training duration jointly.

698 **Larger-scale consistency check.** A separate earlier run at approximately an order of magnitude
 699 larger scale—ViT-Small MoE ($\sim 46\text{M}$ parameters) on full ImageNet-1K for 20 epochs—yields GIR_{lb}
 700 of $0.14\text{--}1.7\times$ across layers at epoch 20 (Layer 3: $1.7\times$, Layer 6: $0.71\times$, Layer 9: $0.14\times$), with
 701 substantial layer-wise variation but no explosion at any layer. This is qualitatively consistent with
 702 the ViT-Tiny ImageNet-50 result ($0.33\text{--}0.51\times$ at Layer 2) reported in Section 4.2, suggesting that
 703 bounded LB GIR in the low-overfitting regime is consistent across substantial variation in both model
 704 scale ($6.86\text{M} \rightarrow \sim 46\text{M}$) and dataset scale (ImageNet-50 $\rightarrow$ ImageNet-1K), even though training
 705 duration, input resolution, and augmentation strategy differ between the two runs.

706 **ViT-Small ImageNet-50 four-condition comparison.** To extend the ViT-Small scale evidence
 707 beyond CIFAR-100 (Section 4.2.3), we train all four conditions on ImageNet-50 with the same
 708 ViT-Small architecture across six seeds ($n=6$; seeds 7, 42, 99, 123, 256, 1337) and apply TOST
 709 equivalence testing ($\delta = \pm 2\text{ pp}$, Dunnett-corrected for $k=3$ many-to-one comparisons against the
 710 shared no-aux baseline).

711 With $n=6$ seeds and Dunnett’s correction ($k=3$, $\rho=0.5$), ERA passes TOST equivalence
 712 ($p_{\text{Dunn}}=0.029$, $\bar{\Delta} = +0.38\text{ pp}$, $d = +0.30$, 90% CI $[-0.65, +1.41]$), suggesting that the accuracy
 713 difference from ERA is within $\pm 2\text{ pp}$. LB ($p_{\text{Dunn}}=0.090$, $\bar{\Delta} = -0.61\text{ pp}$, $d = -0.39$, 90% CI
 714 $[-1.90, +0.67]$) and z-loss ($p_{\text{Dunn}}=0.139$, $\bar{\Delta} = +0.98\text{ pp}$, $d = +0.71$, 90% CI $[-0.15, +2.11]$) fail
 715 Dunnett-corrected TOST: LB shows a mild negative trend within practical tolerance, while z-loss
 716 exhibits a medium positive effect ($d=0.71$), suggesting potential benefit rather than cost but with

Table 11: ViT-Small MoE on ImageNet-50 ($n=6$, final-epoch test accuracy). $\bar{\Delta}$: aux – none (pp). d : Cohen’s d (paired). TOST: $\delta = \pm 2$ pp, paired t -test, Dunnett-corrected ($k=3$ many-to-one comparisons, $\rho=0.5$). GIR from seed = 42 only (epoch 90, layers {3, 6, 9}).

Condition	Mean Acc (%)	$\bar{\Delta}$ (pp)	d	$p_{\text{TOST}}^{\text{Dunn}}$	90% CI	Equiv?	GIR (s42, ep90)
No-Aux	40.03 ± 1.34	—	—	—	—	—	—
LB ($\lambda=0.01$)	39.42 ± 0.47	−0.61	−0.39	0.090	[−1.90, +0.67]	FAIL	0.24–0.54×
ERA ($\lambda=0.1$)	40.41 ± 0.47	+0.38	+0.30	0.029	[−0.65, +1.41]	PASS	0.05–0.10×
Z-loss ($\lambda=0.001$)	41.01 ± 1.48	+0.98	+0.71	0.139	[−0.15, +2.11]	FAIL	1.08–2.34×

high variance across seeds. All four conditions show GIR bounded at $\mathcal{O}(1)$ or below, in contrast to CIFAR-100 where LB and z-loss GIR reach $50\times$ and $438\times$ respectively (Table 15). Z-loss GIR drops from $39.5\text{--}438.8\times$ on CIFAR-100 to $1.08\text{--}2.34\times$ on ImageNet-50—a reduction of $\sim 170\times$ —confirming that regime-dependent GIR dynamics generalize across all three auxiliary loss types at ViT-Small scale. Z-loss GIR remains above $1\times$ at all three layers (L3: $2.34\times$, L6: $1.08\times$, L9: $1.34\times$) yet produces the highest accuracy, further supporting the dissociation between GIR magnitude and accuracy cost.

Leave-one-out sensitivity for ERA equivalence. ERA’s Dunnett-corrected TOST ($p_{\text{Dunn}}=0.029$) warrants robustness assessment; Table 12 reports single-seed exclusion sensitivity.

Table 12: Leave-one-out sensitivity for ViT-Small ImageNet-50 ERA TOST ($n=5$ per exclusion, $\delta = \pm 2$ pp, Dunnett’s $k=3$, $\rho=0.5$).

Excluded Seed	$\bar{\Delta}$ (pp)	SE (pp)	p_{TOST}	$p_{\text{TOST}}^{\text{Dunn}}$	Pass?
7	+0.78	0.40	0.019	0.043	Yes
42	+0.17	0.57	0.016	0.037	Yes
99	+0.08	0.51	0.010	0.022	Yes
123	+0.52	0.60	0.035	0.076	No
256	+0.41	0.63	0.032	0.072	No
1337	+0.33	0.62	0.028	0.060	No

Only 3 of 6 exclusions retain Dunnett significance ($p_{\text{Dunn}} < 0.05$), confirming that ERA equivalence at ViT-Small scale is sensitive to individual seeds. All exclusions maintain $\bar{\Delta}$ within the ± 2 pp bound in absolute terms, and the uncorrected p_{TOST} remains below 0.05 in all cases; the Dunnett failures reflect limited power at $n=5$ rather than directional reversal.

Leave-one-out sensitivity for z-loss. Z-loss on ViT-Small ImageNet-50 exhibits the largest positive mean difference among the four conditions ($\bar{\Delta} = +0.98$ pp at $n=6$), but seed 42 contributes a disproportionate share: the no-aux baseline for seed 42 (39.12%) is anomalously low, inflating Δ_{s42} . Table 13 reports leave-one-out sensitivity across all $n=6$ seeds.

Table 13: Leave-one-out sensitivity for z-loss vs. no-aux on ViT-Small ImageNet-50 ($n=6$ seeds; $\delta = \pm 2$ pp, Dunnett’s $k=3$, $\rho=0.5$). Seed 42 no-aux accuracy (39.12%) is anomalously low; dropping it yields the only Dunnett-significant TOST pass ($p_{\text{Dunn}}=0.009$).

Excluded	n	$\bar{\Delta}$ (pp)	SE (pp)	p_{Dunn}	Pass?
None (full)	6	+0.98	0.56	0.139	No
s7	5	+1.09	0.67	0.248	No
s42	5	+0.48	0.31	0.009	Yes
s99	5	+0.94	0.68	0.200	No
s123	5	+1.24	0.61	0.279	No
s256	5	+0.94	0.69	0.205	No
s1337	5	+1.19	0.64	0.273	No

Only the subset excluding seed 42 passes Dunnett-corrected TOST equivalence ($p_{\text{Dunn}}=0.009$), confirming that the full-sample failure is seed-driven rather than systematic: the anomalously low no-aux accuracy for seed 42 (39.12%) inflates the pairwise Δ and prevents equivalence at $n=6$. All other exclusions yield $\bar{\Delta}$ within ± 1.24 pp and $p_{\text{Dunn}} > 0.20$, indicating insufficient power at $n=5$ rather than directional reversal. The seed 42 pattern is consistent with the ViT-Tiny ImageNet-50 analysis (Table 18), where seed 42 likewise exhibits the largest positive Δ (+3.12 pp) due to an anomalously low baseline (46.32% vs. group mean 48.26%).

Third, ERA’s behavior is generally bounded—at ViT-Tiny scale, the worst-case transient is $1.06\times$, more than an order of magnitude below LB’s sustained explosion. At ViT-Small scale, 6 of 7 seeds yield $\text{GIR}_{\text{era}} \leq 6.72\times$, though one seed (s3) reaches $116.23\times$ via denominator-driven inflation (Section F). Even in this extreme case, the outlier is seed-specific rather than systematic, and the λ -attenuated effective GIR ($\sim 11.6\times$) remains below LB’s sustained $50\times$ regime. This suggests a design principle: objectives that can be approximately *satisfied* (convergent lifecycle) are inherently safer than persistent objectives, though denominator collapse at near-zero task gradients can produce sporadic raw GIR spikes even for convergent losses.

D.5 Medium-Scale Validation: ViT-Base-MoE Table

Table 14: ViT-Base-MoE (~ 98 M) GIR at epoch 80 (ImageNet-50, $n=2$ seeds, $\lambda_{\text{era}}=0.1$). The taxonomy ordering (LB $\gg$ ERA) is consistent across all 8 layer $\times$ seed combinations.

Layer	Seed 42			Seed 123		
	LB GIR	ERA GIR	Ratio	LB GIR	ERA GIR	Ratio
L2	0.77	0.063	$12\times$	0.30	0.028	$11\times$
L5	0.93	0.034	$27\times$	0.90	0.015	$61\times$
L8	0.45	0.009	$50\times$	0.57	0.009	$67\times$
L11	1.03	0.029	$35\times$	0.29	0.010	$29\times$
Mean	0.80	0.034	$24\times$	0.51	0.015	$34\times$

GIV ratio remains within $[0.97, 1.01]$ across all layers and seeds, confirming variance independence at medium scale. Best test accuracy reaches 55.7% (s42) and 54.5% (s123), with a ~ 40 pp train-test gap reflecting the absence of regularization at this scale—a limitation that does not affect GIR analysis (gradient properties are independent of generalization performance).

E Architecture-Dependent Regime Boundaries

The cross-architecture λ sweep reveals three qualitative differences between ViT-Tiny and ConvNeXt-MoE: (i) **Direction reversal at $\lambda=0.001$** : ViT-Tiny achieves $\bar{\Delta} = +1.13$ pp while ConvNeXt-MoE shows $\bar{\Delta} = -0.48$ pp—opposite signs consistent across all 6 seeds, suggesting the ghost-gradient effect is architecture-dependent. (ii) **Seed $\times$ Architecture interaction**: ViT seed 123 is negative in 6/7 λ values, while ConvNeXt seed 123 is positive in 6/7—the same seed index exhibits opposite sensitivity patterns across backbones. (iii) **Stability gap**: ConvNeXt-MoE cross-seed $\text{Std}(\Delta)$ ranges from 0.09 to 0.92 pp across λ values, while ViT-Tiny ranges from 0.87 to 2.92 pp—ConvNeXt consistently exhibits lower variance, possibly due to batch normalization’s implicit regularization of routing gradients. These differences reinforce that the regime boundaries (ghost-gradient, amplifier, stabilizer, saturation) are architecture-dependent; practitioners should not assume λ optima transfer across backbone families. A possible mechanism: L3 cosine alignment remains uniformly near zero ($|\cos \theta| \leq 0.03$) across all 7 λ values for ConvNeXt-MoE, whereas ViT-Tiny L2 cosine oscillates between -0.59 and $+0.09$. This suggests batch normalization randomizes auxiliary gradient direction relative to the task gradient, making GIR magnitude—rather than angular alignment—the primary relevant diagnostic for BN-based architectures.

F ViT-Small CIFAR-100 Detailed Analysis

This appendix provides per-condition analysis for the ViT-Small MoE experiments summarized in Section 4.2.3.

Setup and confounds. The ViT-Small MoE uses 12 transformer blocks with MoE at layers $\{3, 6, 9\}$ (8 experts, top-2 routing), trained for 90 epochs with identical hyperparameters to the ViT-Tiny experiments. At 46.2M parameters, the model is substantially overparameterized for CIFAR-100’s 50k training images (~ 920 parameters per sample vs. ~ 137 for ViT-Tiny), introducing a capacity–data mismatch confound.

Table 15: ViT-Small MoE on CIFAR-100 (90 epochs, $n=3$ seeds). Per-seed test accuracy (%), mean $\pm$ s.d., paired difference $\bar{\Delta}$ (aux – no-aux, pp), paired t -test p -value, and GIR range at epoch 90 across MoE layers $\{3, 6, 9\}$. The 46.2M-parameter model is overparameterized for CIFAR-100 (see text).

Condition	s42	s123	s7	Mean $\pm$ Std	$\bar{\Delta}$ (pp)	p	GIR Range (ep90)
No-Aux	41.73	42.08	38.85	40.89 ± 1.77	—	—	—
LB ($\lambda=0.01$)	37.60	38.62	39.36	38.53 ± 0.88	-2.36	0.245	$4.4\text{--}50.2\times$
ERA ($\lambda=0.1$)	41.20	40.26	42.17	41.21 ± 0.96	$+0.32$	0.854	$0.05\text{--}116.23\times^*$
Z-loss ($\lambda=0.001$)	42.88	41.88	41.66	42.14 ± 0.65	$+1.25$	0.287	$39.5\text{--}438.8\times$

*ERA GIR range over $n=7$ seeds; seed 3 is an extreme outlier ($\text{GIR}_{\text{era}}^{L3}=116.23\times$, denominator-driven; see text).

The no-aux baseline for seed 7 (38.85%) falls ~ 3 pp below the other two seeds (41.73%, 42.08%), causing all three auxiliary conditions to show positive paired differences for that seed and inflating cross-seed variance.

LB: high GIR, high variance, non-significant cost. LB produces GIR of $4.4\text{--}50.2\times$ at epoch 90 (from seed 7: $L3 = 50.19$, $L6 = 25.77$, $L9 = 33.76$), confirming persistent-lifecycle dynamics at $6.7\times$ the ViT-Tiny parameter count. The mean paired accuracy difference is $\bar{\Delta}_{\text{LB}} = -2.36$ pp ($n=3$, $p=0.245$). Per-seed differences exhibit high variance: $\Delta = [-4.13, -3.46, +0.51]$ pp for seeds $\{42, 123, 7\}$.

Routing uniformization. LB produces near-elimination of expert load variance: `load_std` drops 85–87% from baseline ($0.1495 \rightarrow 0.020\text{--}0.022$ across seeds). Routing uniformization may be a necessary but not sufficient condition for accuracy degradation; disentangling this requires larger-scale controlled experiments.

ERA: convergent lifecycle confirmed at scale, with outlier caveat. ERA ($\lambda=0.1$) produces raw GIR of $0.05\text{--}116.23\times$ across seven seeds and three MoE layers. Six of seven seeds yield $\text{GIR}_{\text{era}} \leq 6.72\times$, consistent with the convergent lifecycle observed at ViT-Tiny scale. The exception is seed 3, where Layer 3 GIR_{era} reaches $116.23\times$ —an extreme outlier associated with denominator collapse: the task gradient at L3 approaches near-zero, inflating the raw ratio by three orders of magnitude relative to convergent seeds. This is the same passive amplification mechanism underlying LB GIR explosion (Section 4.1.1), but for ERA it is seed-specific rather than systematic.

The raw GIR range ($0.05\text{--}116.23\times$, spanning three orders of magnitude) overstates the effective interference. After λ -weighting ($\lambda_{\text{era}}=0.1$), the effective ERA contribution remains $< 1\times$ for all convergent seeds; even for seed 3, the λ -attenuated effective GIR at L3 is $\sim 11.6\times$ —large but far below LB’s sustained $50\times$ regime. The dissociation between raw and effective GIR underscores the importance of reporting both metrics: raw GIR reveals denominator-driven inflation, while effective GIR ($= \lambda \cdot \text{GIR}$) captures the actual gradient contribution to the router update. The mean paired accuracy difference is $\bar{\Delta}_{\text{ERA}} = +0.32$ pp ($n=3$, $p=0.854$).

Z-loss: persistent lifecycle, no accuracy cost. Z-loss ($\lambda=0.001$) reveals a persistent lifecycle with GIR of $39.5\text{--}438.8\times$ at epoch 90. Despite this high GIR, $\bar{\Delta}_{\text{Z-loss}} = +1.25$ pp ($n=3$, $p=0.287$). Z-loss preserves expert specialization (`load_std` $\approx 0.17 \pm 0.02$ across seeds (range: $0.154\text{--}0.187$),

comparable to baseline levels ($\approx 0.15\text{--}0.16$)), in contrast to LB’s 85–87% reduction in load variance. High GIR without routing uniformization does not produce an accuracy penalty, suggesting that the interference mechanism, if any, operates through the routing distribution rather than gradient magnitude per se.

G Adam Second-Moment Analysis

To assess whether the optimizer provides additional attenuation beyond λ -weighting, we compare Adam’s second-moment estimates (v_t) and router weight update magnitudes between auxiliary-loss and no-auxiliary-loss conditions at epoch 90, when raw GIR is maximal.

Method. For each router parameter tensor $\theta_R^{(l)}$, we extract the Adam v_t values (running average of squared gradients) from checkpoints at epoch 88 and compute the element-wise ratio $v_t^{\text{aux}}/v_t^{\text{baseline}}$ across matched seeds. We also compute the L2 norm of the weight delta $\|\theta_R^{(90)} - \theta_R^{(88)}\|_2$ as a proxy for the effective update magnitude.

Preliminary findings. Across 4 seeds ($\{42, 99, 123, 256\}$; seeds 7 and 1337 lack epoch-88 checkpoints) and both MoE layers, the Adam v_t ratio (LB vs. baseline) is $1.00\text{--}1.20\times$, far below the $16\text{--}75\times$ raw GIR at the same epoch. Router weight update magnitudes are similarly close: $0.9\text{--}1.2\times$ baseline across layers. These preliminary results suggest that Adam’s per-parameter normalization substantially compresses the raw gradient magnitude imbalance, providing an additional attenuation mechanism beyond the λ -weighting captured by the Gradient Contribution Fraction. However, this analysis covers only the final 2 epochs and a subset of seeds; a full characterization across the training trajectory is left to future work.

H Accuracy Impact Analysis

To quantify whether gradient dominance translates into measurable performance effects, we compare each auxiliary-loss configuration against a no-auxiliary-loss baseline using the same architecture and training protocol.

CIFAR-100 (high overfitting). With $n=6$ seeds per condition, the baseline achieves $45.29 \pm 0.98\%$ best-checkpoint accuracy, load-balancing $45.63 \pm 0.74\%$, and ERA $45.54 \pm 1.04\%$. Paired t -tests find no significant difference for either auxiliary loss (LB: mean diff = $+0.34$ pp, $p = 0.532$, Cohen’s $d = +0.27$; ERA: mean diff = $+0.25$ pp, $p = 0.660$, $d = +0.19$). The 95% confidence intervals span zero in both cases (LB: $[-0.97, +1.66]$; ERA: $[-1.11, +1.61]$), indicating that any accuracy cost of auxiliary-loss gradient interference is indistinguishable from seed variance at this sample size.

ImageNet-50 (low overfitting). With $n=5$ seeds for baseline and ERA, the baseline achieves $48.26 \pm 1.13\%$ test accuracy at epoch 90, and ERA achieves $48.77 \pm 0.31\%$. The paired t -test finds no significant difference (ERA: mean diff = $+0.50$ pp, $p = 0.390$, Cohen’s $d = +0.43$; 95% CI $[-0.95, +1.96]$). For load-balancing ($n=5$), the paired test also shows no significant difference (mean diff = -0.46 pp, $p = 0.444$, $d = -0.38$; 95% CI $[-1.95, +1.04]$). Under uncorrected TOST ($\delta = \pm 2$ pp), all four LB/ERA comparisons reach significance (Table 16). With Dunnett’s correction ($k=3$, $\rho=0.5$), CIFAR-100 LB ($p_{\text{Dunn}}=0.027$) and ERA ($p_{\text{Dunn}}=0.024$) reach significance; ImageNet-50 LB ($p_{\text{Dunn}}=0.051$) and ERA ($p_{\text{Dunn}}=0.051$) narrowly miss, reflecting limited statistical power at $n=5$ rather than meaningful accuracy effects (Table 1).

Z-loss paired comparisons. Table 17 reports per-seed paired accuracy differences for the z-loss condition on CIFAR-100.

Seed 42 shows the largest positive Δ ($+2.24$ pp), slightly exceeding the equivalence bound; the uncorrected TOST is significant ($p = 0.020$) and survives Dunnett’s correction ($p_{\text{Dunn}} = 0.045$).

Z-loss paired comparisons: ImageNet-50. Table 18 reports per-seed paired accuracy differences for z-loss on ImageNet-50, using final-epoch test accuracy.

Table 16: Per-seed paired accuracy differences (pp, aux – none). Positive values indicate aux is higher. Accuracy metric: best checkpoint for CIFAR-100, final epoch for ImageNet-50. [†]Lowest ImageNet-50 baseline seed (see robustness check). Dunnett’s correction for $k=3$ many-to-one comparisons ($\rho=0.5$).

Seed	CIFAR-100		ImageNet-50	
	Δ_{LB}	Δ_{ERA}	Δ_{LB}	Δ_{ERA}
7	+1.38	+0.71	−1.76	+0.48
42 [†]	+0.86	+1.41	+1.24	+2.36
99	−0.21	−2.03	—	—
123	−1.14	−0.20	+0.16	+0.36
256	+1.98	+1.44	−0.56	−0.88
1337	−0.81	+0.15	−1.36	+0.20
$\bar{\Delta}$	+0.34	+0.25	−0.46	+0.50
$p_{t\text{-test}}$	0.532	0.660	0.444	0.390
$p_{TOST} (\delta = \pm 2)$	0.012	0.011	0.023	0.023
$p_{TOST}^{Dunn} (k=3)$	0.027	0.024	0.051	0.051
90% CI (pp)	[−0.69, +1.37]	[−0.82, +1.31]	[−1.60, +0.69]	[−0.61, +1.62]
d	+0.27	+0.19	−0.38	+0.43
n	6	6	5	5

Table 17: Z-loss TOST paired comparisons (ViT-Tiny, CIFAR-100). Peak test accuracy (best checkpoint) reported; CIFAR-100 uses best checkpoint following standard practice for this benchmark, while ImageNet-50 (Table 18) uses final-epoch accuracy as the more conservative metric.

Seed	None (%)	Z-loss (%)	Δ (pp)
42	44.07	46.31	+2.24
123	45.95	46.26	+0.31
7	45.34	45.33	−0.01
99	45.63	44.92	−0.71
1337	46.54	46.89	+0.35
256	44.21	46.16	+1.95
$\bar{\Delta} = +0.69$ pp, SD = 1.16 pp			
$p_{t\text{-test}} = 0.205$; TOST $p = 0.020$ ($\delta = \pm 2$), $p_{Dunn} = 0.045$ ($k=3$), 90% CI = [−0.26, +1.64] pp			

Seed 42 exhibits the largest positive Δ (+3.12 pp), associated with a low baseline (46.32% vs. group mean 48.26%). Grubbs’ test identifies seed 42 as an outlier ($z = 1.90$); however, configuration verification confirms identical hyperparameters across seeds, and the deviation is consistent with seed-specific initialization variance.

Leave-one-out sensitivity. Excluding seed 42 ($n=5$), the mean difference drops to −0.10 pp with TOST $p = 0.0009$, strengthening the equivalence conclusion. Across all six single-seed exclusions (Table 19), $\bar{\Delta}$ ranges from −0.10 pp to +0.70 pp; 3 of 6 exclusions yield $p > 0.05$ (0.051–0.054), reflecting limited statistical power at $n=5$.

Robustness: baseline outlier sensitivity. Seed 42 is the lowest ImageNet-50 baseline performer (46.32% vs. mean 48.26%, -1.72σ). Configuration verification confirms identical hyperparameters to other baseline seeds. Excluding seed 42, the ImageNet-50 ERA paired difference shrinks from +0.50 pp to +0.04 pp ($p = 0.91$, $n=4$), suggesting that the apparent ERA advantage is attributable to a single low-baseline seed rather than a systematic effect. All CIFAR-100 conditions remain non-significant under single-seed exclusion (maximum $|\Delta\bar{\Delta}| = 0.49$ pp, all $p > 0.5$).

Summary. Multi-seed analysis overturns the single-seed observation of 2–3 pp accuracy degradation (which reflected seed-specific variation rather than a systematic effect). The gradient-level effect—real and measurable via GIR and GCF—has not been observed to produce accuracy dif-

Table 18: Z-loss TOST paired comparisons (ViT-Tiny, ImageNet-50). Final epoch test accuracy reported.

Seed	None (%)	Z-loss (%)	Δ (pp)
7	48.60	48.56	−0.04
42	46.32	49.44	+3.12
99	48.16	48.48	+0.32
123	48.36	48.88	+0.52
256	49.20	48.76	−0.44
1337	48.84	47.96	−0.88

$\bar{\Delta} = +0.43$ pp, SD = 1.40 pp
 $p_{\text{t-test}} = 0.486$; TOST $p = 0.021$ ($\delta = \pm 2$), $p_{\text{Dunn}} = 0.047$ ($k=3$), 90% CI = $[-0.73, +1.59]$ pp

Table 19: Leave-one-out sensitivity for ImageNet-50 z-loss TOST ($n=5$ per exclusion, $\delta = \pm 2$ pp).

Excluded Seed	$\bar{\Delta}$ (pp)	SE (pp)	p_{TOST}	Pass?
7	+0.53	0.70	0.051	No
42	−0.10	0.25	< 0.001	Yes
99	+0.46	0.70	0.047	Yes
123	+0.42	0.71	0.044	Yes
256	+0.61	0.67	0.054	No
1337	+0.70	0.63	0.053	No

ferences exceeding ~ 2 pp at micro scale, consistent with the timing mismatch and attenuation mechanisms (λ -weighting, bounded convergence) described in Section 5.1.

I GIR-Aware λ Decay Ablation

We implement a threshold-based λ decay mechanism: at each GIM measurement epoch, if $\max_l \text{GIR}_{\text{lb}}^{(l)} > \tau$, then $\lambda_{\text{lb}} \leftarrow \max(\lambda_{\text{lb}}/2, 10^{-4})$. Table 20 reports paired comparisons against constant- λ baselines on CIFAR-100 ViT-Tiny MoE.

Table 20: GIR-aware λ decay paired comparisons (ViT-Tiny, CIFAR-100). Δ : decay – constant- λ baseline (pp). GIR at epoch 90 (layer 2). Final λ at epoch 90.

τ	Seed	Baseline (%)	Decay (%)	Δ (pp)	$\text{GIR}_{L_2}^{\text{ep90}}$	Final λ
1.0	42	44.74	46.62	+1.88	$389.8 \times$	10^{-4}
1.0	123	44.74	44.83	+0.09	$18.5 \times$	10^{-4}
1.0	256	46.05	45.52	−0.53	$130.0 \times$	10^{-4}
$\tau=1.0$ mean		45.18	45.66	+0.48		
5.0	42	44.74	46.49	+1.75	$11.3 \times$	0.0025
5.0	123	44.74	44.73	−0.01	—	0.00125
5.0	256	46.05	46.16	+0.11	—	0.0025
$\tau=5.0$ mean		45.18	45.79	+0.62		
10.0	42	44.74	46.49	+1.75	$11.3 \times$	0.0025

$\tau=5.0$ and $\tau=10.0$ produce identical results for seed 42 because the GIR phase transition is discontinuous: GIR_{lb} remains at $\sim 1 \times$ through epoch 70, then jumps to $\sim 62 \times$ at epoch 80, triggering decay at the same epoch regardless of threshold. $\tau=1.0$ activates mid-training (epochs 30–50) and reduces λ to its floor (10^{-4}) by epoch 60, but per-seed Δ spans -0.53 to $+1.88$ pp—seed variance exceeds the between-threshold effect.

880 **Three lines of evidence for the timing mismatch.** These results contribute a third line of evidence:
881 **(1) Observational:** effective GIR remains below $0.02\times$ during the post-warmup active learning phase
882 (Section 5.1). **(2) Structural:** the GIR phase transition is discontinuous ($\sim 1\times$ at epoch 70 $\rightarrow \sim 62\times$
883 at epoch 80), so $\tau=5.0$ and $\tau=10.0$ produce *identical* trajectories—any threshold in $[2, 60]$ triggers
884 at the same post-saturation epoch, making reactive decay equivalent to no decay. **(3) Interventional:**
885 even aggressive early decay ($\tau=1.0$), which reduces λ to its floor by epoch 60, fails to yield consistent
886 gains.

887 **GIR rebound caveat.** In a single-seed observation (seed 42; layer-2 $\text{GIR}_{\text{ep90}} = 389.8\times$), $\tau=1.0$
888 produces a GIR rebound far exceeding the constant- λ baseline ($75\times$), though this rebound is not
889 consistent across seeds (seed 123: $18.5\times$, below baseline; seed 256: $130\times$). This suggests that
890 in some training trajectories, aggressive decay may disrupt LB’s regularization of routing weights,
891 though the effect is highly seed-dependent ($n=3$).

892 J Mechanism Analysis: Why eGIR Stays Bounded

893 This section provides the detailed mechanism analysis supporting Section 5.1.

894 The operationally relevant quantity is effective GIR = $\lambda \times \text{GIR}$ (Figure 12 in Appendix C.15). With
895 $\lambda_{\text{lb}} = 0.01$, a raw GIR of $75\times$ translates to $0.75\times$ effective contribution—the maximum across all
896 ViT-Tiny conditions. During active learning (training accuracy $< 99\%$, epochs 5–50), effective GIR
897 stays below $0.02\times$ post-warmup, reaching $0.50 \pm 0.22\times$ only at epoch 90 when accuracy is 99.97%
898 (Figure 10 in Appendix C.13). Two mechanisms attenuate the raw signal: **(1) λ -weighting** compresses
899 LB’s raw GIR by $100\times$; **(2) bounded convergence** independently attenuates ERA ($\sim 35\times$ decay),
900 keeping GIR_{era} below $1\times$ regardless of λ . Adam’s second-moment normalization [Kingma and Ba,
901 2015] may provide a third layer (Appendix G).

902 The observed accuracy differences (ViT-Tiny: $|\bar{\Delta}| \leq 0.69$ pp, $n \geq 5$; ViT-Small: $|\bar{\Delta}| \leq 2.4$ pp, $n=3$,
903 preliminary; Tables 1, 18) coexist with effective GIR < 1 , indicating these costs are not attributable
904 to gradient magnitude. Persistent LB enforces routing uniformization reducing expert specialization
905 ($85\text{--}87\%$ load variance suppression at ViT-Small; Section 4.2.3), while z-loss preserves baseline
906 specialization despite the highest raw GIR ($375\times$, cosine ≈ 0 , $\bar{\Delta} = +0.69$ pp, $p=0.205$ on ViT-Tiny;
907 $+0.01$ pp on ConvNeXt-MoE, $n=3$). This decoupling between gradient-level interference (bounded
908 by λ and direction) and output-level accuracy cost indicates the operative mechanism is distributional—
909 auxiliary losses reshape routing policy—rather than optimization-level gradient competition. The
910 accuracy effect is therefore *loss-type-dependent*: ERA improves accuracy, LB imposes a cost through
911 uniformization, and z-loss is non-significant—consistent across ViT-Tiny ($n \geq 5$) and directionally
912 confirmed on ConvNeXt-MoE ($n=3$). A pilot λ -decay ablation (Appendix I) confirms the timing
913 mismatch: reactive decay is equivalent to no decay because the GIR phase transition occurs only
914 post-saturation.

915 K Extended Practical Implications

916 This section provides the detailed regime analysis and dose-response discussion supporting Sec-
917 tion 5.2.

918 Four behavioral regimes emerge from the ViT-Tiny λ sweep (Section 4.1.2): ghost-gradient ($\lambda=0.001$,
919 GIR explosion with near-zero cosine—no accuracy damage on ViT-Tiny; mildly negative on
920 ConvNeXt-MoE (Appendix C.7)), amplifier ($\lambda=0.01$, seed-specific collapse), stabilizer ($\lambda=0.2$, opti-
921 mal mean and reduced variance), and saturation ($\lambda \geq 0.5$, diminishing returns). The ghost-gradient vs.
922 amplifier distinction provides the cleanest evidence that GIR magnitude alone is insufficient—angular
923 alignment determines whether GIR inflation translates to accuracy damage. A ConvNeXt-MoE
924 sweep ($n=3$; Appendix C.7) confirms the self-limiting property extends cross-architecture, with L3
925 GIR monotonically decreasing from $\sim 16\times$ ($\lambda=0.001$) to $0.30\times$ ($\lambda=1.0$). Persistent losses such as
926 load-balancing lack this self-limiting feedback: their gradients do not vanish with stronger λ , so the
927 “higher $\lambda \Rightarrow$ higher risk” concern remains valid for them.

928 The non-monotonic λ –GIR relationship admits *dose-response saturation*: higher λ amplifies the ERA
929 gradient early, accelerating router alignment, which accompanies ERA loss approaching zero—so
930 $\lambda \cdot \|\nabla \mathcal{L}_{\text{era}}\|$ decreases despite the larger coefficient. Late-training GIR > 1 must distinguish active

interference (auxiliary gradient genuinely large and aligned) from passive inflation ($\|\mathbf{g}_{\text{task}}\| \rightarrow 0$; Appendix D). The ghost-gradient regime is pure passive inflation; the amplifier regime mixes both. GIM serves as a diagnostic tool: a rising post-warmup GIR trend signals approaching gradient dominance *before* it affects accuracy, enabling informed λ decay or loss-free balancing [Wang et al., 2024, DeepSeek-AI, 2024, Han and Zhong, 2025].

GIM as diagnostic, not prescriptive. GIM metrics (GIR, cosine similarity) characterize gradient interaction dynamics but do not serve as accuracy estimators. Regime classification (Section 4.1.2) relies on accuracy response patterns (broadly increasing, inverted-U, collapse) and GIR directional trends, not on cosine absolute values. ConvNeXt-MoE at $\lambda=1.0$ illustrates this decoupling: L2 cosine similarity varies widely across seeds ($-0.41, -0.78, -0.88$ for seeds 42, 123, 256; Appendix C.7), yet accuracy remains stable ($\sigma=0.09$ pp). This seed variability reflects batch normalization’s tolerance to gradient interference—even deeply opposing cosine (-0.88) is absorbed without accuracy degradation. The deeper L3 layer corroborates attenuation through depth, with narrower cosine spread ($+0.01, -0.06, -0.04$ across the same seeds).

L Extended Limitations Discussion

This section provides additional limitation details supporting Section 5.3.

Auxiliary losses and optimizer. We study three auxiliary losses (load-balancing, ERA, router z-loss); other routing regularizers—expert choice [Zhou et al., 2022], hash routing, or loss-free balancing [Wang et al., 2024, DeepSeek-AI, 2024, Han and Zhong, 2025]—are not evaluated, and the z-loss vs. LB comparison is confounded by differing λ values. All experiments use AdamW [Loshchilov and Hutter, 2019]; Adam’s second-moment normalization may provide implicit gradient attenuation (Appendix G), so optimizers without adaptive second moments could exhibit different effective-GIR dynamics.

Detailed statistical power analysis. The equivalence margin $\delta=\pm 2$ pp was fixed *a priori*, informed by baseline standard deviation $\sigma \approx 1$ pp, representing a practically meaningful effect size ($\approx 2\sigma$). We deliberately avoid adaptive equivalence bounds; the fixed margin ensures comparability across conditions and prevents post hoc threshold selection. TOST equivalence testing [Schuirmann, 1987, Lakens, 2017] is applied with Dunnett’s many-to-one correction [Dunnett, 1955] ($k=3, \rho=0.5$), using $n=3$ seeds per condition in λ -sweep experiments and $n=4-6$ per condition in regime ablations. Under a two-sample t -test with $\sigma=1$ pp and $\delta=2$ pp, statistical power at $n=3$ is approximately 0.3–0.4; regime ablations with $n=4-6$ reach modestly higher power but remain limited. Null TOST results are therefore inconclusive—insufficient evidence to conclude equivalence—rather than evidence of no effect. The ERA stabilizer effect is seed-dependent (s123 yields negative Δ in 6 of 7 λ values; Section 4.1.2), underscoring the need for larger-scale replication.

Equicorrelation sensitivity (ρ). Dunnett’s correction assumes equicorrelation ρ among treatments sharing a common control. We use $\rho=0.5$ (the theoretical value for balanced paired designs); the empirical mean correlation across CIFAR-100 seed-matched deltas is $\bar{\rho}=0.381$. Table 21 reports the ρ -sensitivity of ImageNet-50 z-loss, the only condition whose PASS/FAIL status depends on this assumption. All three CIFAR-100 conditions remain robust across all ρ values tested ($p_{\text{Dunn}} < 0.05$ at $\rho=0.30$).

GIM measurement granularity. GIM diagnostics are computed from a single training batch per measurement epoch, yielding point estimates without within-epoch confidence intervals. While consistent with common practice for gradient-level diagnostics [Chen et al., 2018, Yu et al., 2020], batch-level stochasticity may introduce noise, particularly in early training. Cross-seed replication captures between-run variability but not within-epoch measurement noise; multi-batch estimation could reduce this limitation at additional computational cost.

M Future Work

GIR-aware λ scheduling. The U-shape trajectory (Section 4.1.1) motivates threshold-based λ decay: halving λ_{lb} when GIR exceeds τ . Our ablation (Section I; Table 20) shows a flat dose-response

Table 21: ρ -sensitivity of Dunnett-corrected TOST for z-loss ($k=3$, $\delta=\pm 2$ pp, MC 5×10^6 samples). Each row uses equicorrelation at the specified ρ . IN50 z-loss: $n=6$, $df=4$ (block min). CIFAR z-loss: $n=6$, $df=5$.

ρ	IN50 z-loss p_{Dunn}	CIFAR z-loss p_{Dunn}	Status
0.30	0.0513	0.0494	IN50 FAIL; CIFAR PASS
0.35	0.0501	0.0483	IN50 FAIL; CIFAR PASS
0.381*	0.0497	0.0475	IN50 PASS (margin 0.0003)
0.40	0.0491	0.0472	IN50 PASS
0.50	0.0468	0.0447	IN50 PASS

* Empirical mean correlation $\bar{\rho}$ across CIFAR-100 seed-matched deltas.

across $\tau \in \{1.0, 5.0, 10.0\}$, consistent with the timing mismatch: by the time GIR rises, training has saturated. Whether GIR-aware scheduling provides benefits at larger scales—where task gradient collapse may occur during active learning rather than post-saturation—remains open. GradNorm-style [Chen et al., 2018] continuous adaptation, triggered by gradient *ratio* dynamics rather than loss-scale normalization, is a natural extension.

Further scale validation. Our ViT-Base-MoE (~ 98 M) experiments (Section 4.2.5) confirm taxonomy ordering at medium scale (at $\lambda_{\text{era}}=0.1$), but the next frontier is ViT-Large/Huge MoE (300M+ parameters) trained for hundreds of epochs on full ImageNet [Riquelme et al., 2021], where models may transition between overfitting regimes within a single run. The larger number of MoE layers and experts in scaled models introduces more router parameters subject to auxiliary-loss gradient interference, possibly amplifying cumulative effects. GIM is computationally cheap and can be applied to production training runs without significant overhead.

Continual learning and fine-tuning. The accuracy cost of auxiliary-loss interference may compound under distribution shift. When a pre-trained MoE is fine-tuned on a downstream task, the initial task gradient is large but may collapse quickly as the model adapts, triggering rapid GIR explosion analogous to the late-training regime we observe. GIM trajectories during fine-tuning could reveal whether auxiliary losses should be decayed or disabled during adaptation, complementing existing work on loss weighting in multi-task settings [Kendall et al., 2018, Navon et al., 2022].

Router architecture interaction. Our experiments use linear routers. Different router designs—MLP routers, hash-based routing, soft routing [Puigcerver et al., 2024]—have qualitatively different gradient flow properties. GIM could reveal whether the differential convergence pattern is specific to linear routers (where the gradient landscape is simpler) or a universal property of load-balancing objectives. Hash-based routers, which have no learnable routing parameters, would eliminate auxiliary-loss gradient interference entirely but at the cost of adaptivity.

Extension to language MoE. Language model training runs are typically much longer than our 90-epoch setting, and the task objective (next-token prediction on diverse corpora) may sustain a non-trivial gradient throughout training. The accuracy cost could manifest differently: rather than a late-training explosion, it might produce sustained moderate interference that accumulates over hundreds of thousands of steps. Switch Transformer [Fedus et al., 2022] and GShard [Lepikhin et al., 2021] both use load-balancing losses with fixed coefficients; GIM analysis could determine whether these impose hidden costs on routing quality.

N Broader Impact

GIM as a general diagnostic. Beyond this specific study, GIM provides a lightweight, model-agnostic diagnostic for any multi-loss training setup—not limited to MoE. Any system that combines a primary objective with auxiliary losses (multi-task learning [Yu et al., 2020, Liu et al., 2021], regularized training, self-supervised pre-training with auxiliary heads) can exhibit gradient dominance dynamics that GIR trajectories would reveal. Code for GIM computation and reproduction of all experiments will be made available upon publication.

Rethinking static loss coefficients. The differential convergence pattern implies that the optimal λ value is training-phase-dependent: the raw auxiliary-to-task gradient ratio grows by nearly two orders of magnitude (from $\sim 1\times$ at epoch 30 to $\sim 75\times$ at epoch 90 in the controlled ablation), though the λ -weighted effective contribution shifts more modestly (task GCF declining from 72% to 42%). This challenges the standard practice of setting auxiliary loss coefficients via grid search at a single evaluation point and keeping them fixed. Our findings motivate adaptive scheduling as a default consideration rather than an optimization afterthought, echoing similar arguments in multi-task learning [Chen et al., 2018, Kendall et al., 2018].

Implications for accessible MoE. As MoE architectures become viable at smaller scales—for edge devices, academic research, and resource-constrained settings [Zhou et al., 2022]—understanding auxiliary loss dynamics becomes critical. The accuracy cost of gradient interference is likely *more* severe at micro scale, where the task gradient signal is scarce and converges faster, leaving auxiliary losses to dominate a larger fraction of training. The raw GIR_{lb} values of up to $110\times$ we observe at 6.86M parameters (effective contribution $0.50 \pm 0.22\times$ after λ -weighting) may attenuate at larger scales where the task gradient is better sustained. Practitioners working at small scale should monitor GIR trajectories and consider λ decay schedules, particularly for persistent auxiliary objectives like load balancing.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: **Yes.**

Justification: The abstract and introduction (Section 1) state two contributions—GIM diagnostic framework and empirical characterization of the accuracy cost (differential convergence pattern, timing mismatch, regime ablation)—each supported by corresponding experimental results in Sections 4.1–4.2. Scope limitations (micro-scale, three auxiliary losses, single architecture) are explicitly acknowledged in Section 5.3.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: **Yes.**

Justification: Section 5.3 discusses limitations including micro-scale scope, statistical power ($n=5-6$ seeds), restriction to three auxiliary losses with λ confounding, dataset-regime confounding, preliminary optimizer interaction analysis, and absence of loss-free balancing comparison.

3. Theory, Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

Answer: **N/A.**

Justification: This is an empirical study. The GIM framework (Section 3) defines diagnostic quantities (GIR, AA, GCF) but makes no theoretical claims requiring proof. The differential convergence distinction is mathematically expected; our contribution is quantitative characterization.

4. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

Answer: **Yes.**

Justification: Section 3.5 specifies architecture (ViT-Tiny MoE, 6.86M parameters), auxiliary loss formulations, hyperparameters (learning rate, weight decay, batch size, augmentation), training protocol (epochs, seeds, warmup), and GIM measurement procedure. Appendix B provides additional architectural and measurement details.

1069 **5. Open Access to Data and Code**

1070 Question: Does the paper provide open access to the data and code, with sufficient instruc-
1071 tions to faithfully reproduce the main experimental results?

1072 Answer: **Yes.**

1073 Justification: Code for GIM computation and all experiments will be released upon publica-
1074 tion (Appendix N). Experiments use publicly available datasets (CIFAR-100, ImageNet-1K
1075 subset).

1076 **6. Experimental Setting/Details**

1077 Question: Does the paper specify all the training and test details necessary to understand the
1078 results?

1079 Answer: **Yes.**

1080 Justification: Section 3.5 reports architecture, optimizer (AdamW), learning rate schedule
1081 (cosine annealing with 5-epoch warmup), batch size (128), augmentation (RandomCrop,
1082 RandomHorizontalFlip, Normalize), hardware (4× RTX PRO 6000), seeds, and GIM
1083 measurement intervals.

1084 **7. Experiment Statistical Significance**

1085 Question: Does the paper report error bars suitably and correctly defined or other appropriate
1086 information about the statistical significance of the experiments?

1087 Answer: **Yes.** TOST equivalence testing with $\Delta = \pm 2$ pp boundary; Dunnett’s correction
1088 discussed in Section 5.3.

1089 Justification: All GIR results report ranges across seeds (e.g., GIR_{lb} 2.8–75× across 6 seeds).
1090 Table 1 reports mean $\pm$ s.d. test accuracy with paired t -test p -values ($n=5-6$ per condition).
1091 Figure 3 displays seed-range shaded bands. Section 5.3 acknowledges that GIR magnitudes
1092 are seed-sensitive (one seed exhibited router collapse).

1093 **8. Experiments Compute Resource**

1094 Question: For each experiment, does the paper provide sufficient information on the com-
1095 puter resources needed to reproduce the experiments?

1096 Answer: **Yes.**

1097 Justification: Section 3.5 specifies 4× RTX PRO 6000 GPUs. Each 90-epoch CIFAR-100
1098 run takes approximately 1 hour; ImageNet-50 runs take approximately 2.5 hours. Total
1099 compute for all experiments is under 100 GPU-hours.

1100 **9. Code Of Ethics**

1101 Question: Does the research conform to the NeurIPS Code of Ethics?

1102 Answer: **Yes.**

1103 Justification: This work introduces a diagnostic framework for understanding gradient
1104 dynamics in MoE training. It does not involve human subjects, personal data, or dual-use
1105 concerns.

1106 **10. Broader Impacts**

1107 Question: Does the paper discuss both potential positive societal impacts and negative
1108 societal impacts of the work?

1109 Answer: **Yes.**

1110 Justification: Appendix N discusses GIM as a general diagnostic tool, implications for
1111 rethinking static loss coefficients, and relevance for accessible MoE at smaller scales.

1112 **11. Safeguards**

1113 Question: Does the paper describe safeguards that have been put in place for responsible
1114 release of data or models with a high risk for misuse?

1115 Answer: **N/A.**

1116 Justification: This work releases a diagnostic framework (GIM) and micro-scale experimen-
1117 tal code. Neither poses misuse risks.

1118 **12. Licenses for Existing Assets**

1119 Question: Are the creators of assets used in the paper properly credited and are the license

1120 and terms of use explicitly mentioned and properly respected?

1121 Answer: **Yes.**

1122 Justification: CIFAR-100 and ImageNet are standard public benchmarks. The ViT architec-

1123 ture follows [Dosovitskiy et al. \[2021\]](#). The load-balancing loss follows [Fedus et al. \[2022\]](#).

1124 All sources are properly cited.

1125 **13. New Assets**

1126 Question: Are new assets introduced in the paper well documented and is the documentation

1127 provided alongside the assets?

1128 Answer: **Yes.**

1129 Justification: The GIM framework is fully documented in Section 3 with equations and

1130 measurement protocol. Code will be released with documentation upon publication.

1131 **14. Crowdsourcing and Research with Human Subjects**

1132 Question: For crowdsourcing experiments and research with human subjects, does the paper

1133 include the full text of instructions given to participants and screenshots?

1134 Answer: **N/A.**

1135 Justification: This work does not involve crowdsourcing or human subjects.

1136 **15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human**

1137 **Subjects**

1138 Question: Does the paper describe potential risks and limitations of the research to human

1139 subjects?

1140 Answer: **N/A.**

1141 Justification: This work does not involve human subjects.

1142 **16. Declaration of LLM usage**

1143 Question: Does the paper describe the usage of LLMs if it is an important, original compo-

1144 nent of the core methods?

1145 Answer: **N/A.**

1146 Justification: LLMs are not a component of the proposed GIM framework or experimental

1147 methodology. LLM-assisted tools were used for manuscript preparation but not for any

1148 experimental results or scientific claims.